

Theory of Algebraic Topology 17

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Lecture notes with worked examples in algebraic topology

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1 Theory for Problems 8 and 9

1.1 Problem 8: Acyclic free chain complexes and chain contractions

Statement. Suppose that $(C_\bullet, d)$ is a chain complex of finitely generated free abelian groups such that

$$H_*(C_\bullet, d) = 0.$$

Show that

$$\text{id}: C_\bullet \rightarrow C_\bullet$$

is chain homotopic to the zero map.

Chain complexes

A **chain complex** is a sequence of abelian groups and homomorphisms

$$\cdots \xrightarrow{d_{n+2}} C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \xrightarrow{d_{n-1}} \cdots$$

such that

$$d_n \circ d_{n+1} = 0$$

for every n . This condition means that every boundary is automatically a cycle.

For each n , we define the group of n -cycles by

$$Z_n(C) = \ker d_n$$

and the group of n -boundaries by

$$B_n(C) = \text{im } d_{n+1}.$$

Since $d_n d_{n+1} = 0$, we have

$$B_n(C) \subseteq Z_n(C).$$

The n -th homology group of the chain complex is

$$H_n(C) = \frac{Z_n(C)}{B_n(C)} = \frac{\ker d_n}{\text{im } d_{n+1}}.$$

Thus $H_n(C)$ measures the difference between cycles and boundaries. If $H_n(C) = 0$, then every n -cycle is an n -boundary.

Acyclic chain complexes

The condition

$$H_*(C_\bullet, d) = 0$$

means that

$$H_n(C_\bullet, d) = 0$$

for every n . Therefore,

$$Z_n(C) = B_n(C)$$

for every n , or equivalently,

$$\ker d_n = \operatorname{im} d_{n+1}.$$

Hence the chain complex is exact at every degree:

$$\cdots \rightarrow C_{n+1} \rightarrow C_n \rightarrow C_{n-1} \rightarrow \cdots .$$

Such a chain complex is called **acyclic**. Intuitively, acyclicity means that the complex has no homological holes.

Chain maps and chain homotopies

A chain map

$$f: C_\bullet \rightarrow C_\bullet$$

is a collection of homomorphisms

$$f_n: C_n \rightarrow C_n$$

such that

$$d_n f_n = f_{n-1} d_n$$

for every n .

Two chain maps

$$f, g: C_\bullet \rightarrow C_\bullet$$

are called **chain homotopic** if there exist homomorphisms

$$h_n: C_n \rightarrow C_{n+1}$$

such that

$$f_n - g_n = d_{n+1} h_n + h_{n-1} d_n$$

for every n .

In particular, saying that the identity map is chain homotopic to the zero map means that there exist homomorphisms

$$h_n: C_n \rightarrow C_{n+1}$$

such that

$$\operatorname{id}_{C_n} = d_{n+1} h_n + h_{n-1} d_n$$

for every n .

A chain complex satisfying this condition is called **chain-contractible**.

Why freeness is important

Each C_n is assumed to be a finitely generated free abelian group. Thus

$$C_n \cong \mathbb{Z}^{r_n}$$

for some integer $r_n \geq 0$.

A subgroup of a free abelian group is again free abelian. Therefore the boundary groups

$$B_n(C) = \text{im } d_{n+1}$$

are free abelian groups.

Since the complex is acyclic, we have

$$B_n(C) = Z_n(C) = \ker d_n.$$

Therefore, for each n , we get a short exact sequence

$$0 \rightarrow B_n(C) \rightarrow C_n \xrightarrow{d_n} B_{n-1}(C) \rightarrow 0.$$

The group $B_{n-1}(C)$ is free abelian, hence projective. Therefore this short exact sequence splits. Consequently, we may write

$$C_n \cong B_n(C) \oplus A_n$$

for some subgroup $A_n \subseteq C_n$.

Moreover, the differential restricts to an isomorphism

$$d_n|_{A_n} : A_n \rightarrow B_{n-1}(C).$$

Indeed, the kernel of d_n is $B_n(C)$, and A_n is chosen as a complement to $B_n(C)$ inside C_n . Thus d_n is injective on A_n , and its image is $B_{n-1}(C)$.

Construction of the chain homotopy

Using the splitting

$$C_n = B_n(C) \oplus A_n,$$

every element $c \in C_n$ can be written uniquely as

$$c = b + a,$$

where

$$b \in B_n(C), \quad a \in A_n.$$

Since

$$d_{n+1}|_{A_{n+1}} : A_{n+1} \rightarrow B_n(C)$$

is an isomorphism, we define

$$h_n : C_n \rightarrow C_{n+1}$$

by

$$h_n(b + a) = \left(d_{n+1}|_{A_{n+1}}\right)^{-1}(b).$$

Now compute:

$$d_{n+1}h_n(b + a) = b.$$

Also,

$$d_n(b + a) = d_n(b) + d_n(a).$$

Since $b \in B_n(C) = \ker d_n$, we have $d_n(b) = 0$. Therefore

$$d_n(b + a) = d_n(a).$$

But

$$d_n(a) \in B_{n-1}(C).$$

Hence

$$h_{n-1}d_n(b + a) = h_{n-1}(d_n(a)) = a.$$

Therefore,

$$d_{n+1}h_n(b + a) + h_{n-1}d_n(b + a) = b + a.$$

So

$$d_{n+1}h_n + h_{n-1}d_n = \text{id}_{C_n}.$$

Thus

$$\text{id}_{C_\bullet} \simeq 0.$$

Therefore the chain complex $C_\bullet$ is chain-contractible.

Basic example

Consider the chain complex

$$0 \rightarrow \mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z} \rightarrow 0.$$

Here

$$C_1 = \mathbb{Z}, \quad C_0 = \mathbb{Z}, \quad d_1 = \text{id}.$$

We have

$$H_1(C) = 0$$

because

$$\ker d_1 = 0.$$

Also,

$$H_0(C) = 0$$

because

$$\text{im } d_1 = \mathbb{Z} = C_0.$$

Hence the complex is acyclic. Define

$$h_0: C_0 \rightarrow C_1$$

by

$$h_0 = \text{id}_{\mathbb{Z}}.$$

Then

$$d_1 h_0 = \text{id}_{C_0}.$$

This is the simplest example of a contraction. The general proof above says that every acyclic chain complex of finitely generated free abelian groups decomposes into such elementary cancellation pieces.

1.2 Problem 9: Products of polyhedra of simplicial complexes

Statement. For simplicial complexes K and L inside $\mathbb{R}^m$ and $\mathbb{R}^n$, respectively, show that

$$|K| \times |L| \subset \mathbb{R}^{m+n} = \mathbb{R}^m \times \mathbb{R}^n$$

is the polyhedron of a simplicial complex.

Simplices

Let $v_0, \dots, v_p \in \mathbb{R}^m$ be affinely independent points. The simplex spanned by these points is

$$[v_0, \dots, v_p] = \left\{ \sum_{i=0}^p t_i v_i \mid t_i \geq 0, \sum_{i=0}^p t_i = 1 \right\}.$$

Examples:

$$[v_0]$$

is a point,

$$[v_0, v_1]$$

is an interval,

$$[v_0, v_1, v_2]$$

is a triangle, and

$$[v_0, v_1, v_2, v_3]$$

is a tetrahedron.

A **face** of a simplex is obtained by taking a subset of its vertices. For example, the faces of a triangle are its three vertices, its three edges, and the triangle itself.

Simplicial complexes

A **simplicial complex** K is a collection of simplices such that:

$$\text{if } \sigma \in K \text{ and } \tau \leq \sigma, \text{ then } \tau \in K,$$

and

$$\sigma \cap \tau$$

is either empty or a common face of both σ and τ .

The **polyhedron** of K , denoted $|K|$, is the union of all simplices in K :

$$|K| = \bigcup_{\sigma \in K} \sigma.$$

Thus $|K|$ is the geometric space represented by the simplicial complex.

The product problem

Suppose

$$K \subset \mathbb{R}^m, \quad L \subset \mathbb{R}^n.$$

Then

$$|K| \subset \mathbb{R}^m, \quad |L| \subset \mathbb{R}^n.$$

Therefore

$$|K| \times |L| \subset \mathbb{R}^m \times \mathbb{R}^n = \mathbb{R}^{m+n}.$$

We want to prove that $|K| \times |L|$ is itself the polyhedron of some simplicial complex.

The key idea is:

$$|K| \times |L| = \bigcup_{\sigma \in K, \tau \in L} \sigma \times \tau.$$

So it is enough to understand how to triangulate products of simplices.

First case: product of two simplices

Let

$$\sigma = \Delta^p = [v_0, \dots, v_p]$$

and

$$\tau = \Delta^q = [w_0, \dots, w_q].$$

Then

$$\sigma \times \tau = \Delta^p \times \Delta^q.$$

The product of two simplices is usually not a simplex. For example,

$$\Delta^1 \times \Delta^1$$

is a square, not a simplex. However, it can be triangulated into two triangles.

Similarly,

$$\Delta^2 \times \Delta^1$$

is a triangular prism, which can be triangulated into three tetrahedra.

In general, $\Delta^p \times \Delta^q$ can be triangulated by the **staircase triangulation**.

The staircase triangulation

The vertices of

$$\Delta^p \times \Delta^q$$

are pairs

$$(v_i, w_j), \quad 0 \leq i \leq p, \quad 0 \leq j \leq q.$$

A maximal simplex in the staircase triangulation corresponds to a monotone lattice path from

$$(0, 0)$$

to

$$(p, q).$$

Such a path uses only steps of the form

$$(i, j) \mapsto (i + 1, j)$$

and

$$(i, j) \mapsto (i, j + 1).$$

A path has $p + q$ steps and therefore visits $p + q + 1$ lattice points. These lattice points determine $p + q + 1$ vertices of $\Delta^p \times \Delta^q$, hence a simplex of dimension $p + q$.

For example, a path

$$(0, 0) = (i_0, j_0), (i_1, j_1), \dots, (i_{p+q}, j_{p+q}) = (p, q)$$

determines the simplex

$$[(v_{i_0}, w_{j_0}), \dots, (v_{i_{p+q}}, w_{j_{p+q}})].$$

The number of maximal simplices is the number of ways to choose where the p horizontal steps occur among $p + q$ total steps:

$$\binom{p+q}{p} = \binom{p+q}{q}.$$

Hence

$$\Delta^p \times \Delta^q$$

can be triangulated into

$$\binom{p+q}{p}$$

simplices of dimension $p + q$.

Example: square as a product of intervals

Let

$$\Delta^1 = [v_0, v_1], \quad \Delta^1 = [w_0, w_1].$$

Then

$$\Delta^1 \times \Delta^1$$

has vertices

$$(v_0, w_0), \quad (v_1, w_0), \quad (v_0, w_1), \quad (v_1, w_1).$$

Geometrically, this is a square.

There are two monotone paths from $(0, 0)$ to $(1, 1)$:

$$(0, 0) \rightarrow (1, 0) \rightarrow (1, 1)$$

and

$$(0, 0) \rightarrow (0, 1) \rightarrow (1, 1).$$

They give two triangles:

$$[(v_0, w_0), (v_1, w_0), (v_1, w_1)]$$

and

$$[(v_0, w_0), (v_0, w_1), (v_1, w_1)].$$

Thus the square is triangulated into two triangles.

Example: triangular prism

Let

$$\Delta^2 = [v_0, v_1, v_2], \quad \Delta^1 = [w_0, w_1].$$

Then

$$\Delta^2 \times \Delta^1$$

is a triangular prism.

The staircase triangulation has

$$\binom{2+1}{2} = 3$$

maximal simplices. Therefore the triangular prism is triangulated into three tetrahedra.

Compatibility with faces

The crucial point is that the staircase triangulation is compatible with faces.

If

$$\sigma' \leq \sigma$$

and

$$\tau' \leq \tau,$$

then

$$\sigma' \times \tau'$$

is a face-like subproduct of

$$\sigma \times \tau.$$

The staircase triangulation of

$$\sigma' \times \tau'$$

is exactly the restriction of the staircase triangulation of

$$\sigma \times \tau.$$

This compatibility is what allows the local triangulations of each product

$$\sigma \times \tau$$

to glue together into one global simplicial complex.

The general case

Now let K and L be arbitrary simplicial complexes. Choose an ordering of the vertices of K and an ordering of the vertices of L . For each pair of simplices

$$\sigma = [v_0, \dots, v_p] \in K$$

and

$$\tau = [w_0, \dots, w_q] \in L,$$

triangulate

$$\sigma \times \tau$$

using the staircase triangulation.

Let $K \boxtimes L$ denote the collection of all simplices obtained in this way, for all pairs

$$\sigma \in K, \quad \tau \in L.$$

Because the staircase triangulations are compatible on common faces, the collection $K \boxtimes L$ is a simplicial complex.

Its polyhedron is

$$|K \boxtimes L| = \bigcup_{\sigma \in K, \tau \in L} (\sigma \times \tau).$$

But

$$\bigcup_{\sigma \in K, \tau \in L} (\sigma \times \tau) = \left(\bigcup_{\sigma \in K} \sigma \right) \times \left(\bigcup_{\tau \in L} \tau \right).$$

Therefore

$$|K \boxtimes L| = |K| \times |L|.$$

Hence

$$|K| \times |L|$$

is the polyhedron of a simplicial complex.

Main conclusion

The product of two simplices is usually not itself a simplex, but it always has a natural triangulation. The staircase triangulation gives

$$\Delta^p \times \Delta^q$$

as a union of

$$\binom{p+q}{p}$$

simplices of dimension $p+q$.

Since every simplicial complex is built out of simplices, we triangulate each product

$$\sigma \times \tau, \quad \sigma \in K, \quad \tau \in L,$$

and the compatibility of the staircase triangulation along faces guarantees that these local triangulations glue together correctly.

Thus

$$|K| \times |L| \subset \mathbb{R}^{m+n}$$

is the polyhedron of a simplicial complex.

2 Examples for Problems 8 and 9

2.1 Examples for Problem 8: Acyclic Free Chain Complexes

Recall the statement:

$$(C_\bullet, d)$$

is a chain complex of finitely generated free abelian groups and

$$H_*(C_\bullet, d) = 0.$$

Then the identity chain map

$$\text{id}_{C_\bullet}: C_\bullet \rightarrow C_\bullet$$

is chain homotopic to the zero map. Equivalently, there exist homomorphisms

$$h_n: C_n \rightarrow C_{n+1}$$

such that

$$\text{id}_{C_n} = d_{n+1}h_n + h_{n-1}d_n$$

for every n .

Example 8.1: The simplest contractible chain complex

Consider the chain complex

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z} \longrightarrow 0.$$

Here

$$C_1 = \mathbb{Z}, \quad C_0 = \mathbb{Z}, \quad d_1 = \text{id}_{\mathbb{Z}}.$$

We compute the homology.

First,

$$H_1(C) = \ker d_1.$$

Since

$$d_1 = \text{id}_{\mathbb{Z}},$$

we have

$$\ker d_1 = 0.$$

Therefore

$$H_1(C) = 0.$$

Next,

$$H_0(C) = \frac{C_0}{\text{im } d_1}.$$

Since

$$\text{im } d_1 = \mathbb{Z},$$

we get

$$H_0(C) = \frac{\mathbb{Z}}{\mathbb{Z}} = 0.$$

Hence the complex is acyclic.

Now define

$$h_0: C_0 \rightarrow C_1$$

by

$$h_0 = \text{id}_{\mathbb{Z}}.$$

Then on C_0 ,

$$d_1 h_0 = \text{id}_{C_0}.$$

On C_1 , the chain homotopy equation becomes

$$h_0 d_1 = \text{id}_{C_1}.$$

Thus

$$\text{id}_{C_\bullet} \simeq 0.$$

This is the basic cancellation pair:

$$\mathbb{Z} \xrightarrow{\cong} \mathbb{Z}.$$

Example 8.2: A complex that is not acyclic

Consider

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow 0.$$

Here

$$d_1(n) = 2n.$$

We compute the homology.

First,

$$H_1(C) = \ker d_1.$$

Multiplication by 2 is injective, so

$$\ker d_1 = 0.$$

Thus

$$H_1(C) = 0.$$

However,

$$H_0(C) = \frac{C_0}{\operatorname{im} d_1} = \frac{\mathbb{Z}}{2\mathbb{Z}}.$$

Hence

$$H_0(C) \cong \mathbb{Z}/2\mathbb{Z} \neq 0.$$

Therefore this complex is not acyclic.

This example shows that injectivity of the boundary map is not enough. We also need surjectivity at the next degree. The map

$$\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}$$

does not hit the odd integers, so nonzero homology remains.

Example 8.3: A three-term exact complex

Consider the chain complex

$$0 \longrightarrow \mathbb{Z} \xrightarrow{d_2} \mathbb{Z}^2 \xrightarrow{d_1} \mathbb{Z} \longrightarrow 0,$$

where

$$d_2(n) = (n, n)$$

and

$$d_1(a, b) = a - b.$$

First we check that this is a chain complex:

$$d_1 d_2(n) = d_1(n, n) = n - n = 0.$$

Therefore

$$d_1 d_2 = 0.$$

Now compute the homology.

At degree 2,

$$H_2(C) = \ker d_2.$$

If

$$d_2(n) = 0,$$

then

$$(n, n) = (0, 0),$$

so

$$n = 0.$$

Therefore

$$H_2(C) = 0.$$

At degree 1,

$$H_1(C) = \frac{\ker d_1}{\operatorname{im} d_2}.$$

We have

$$d_1(a, b) = a - b.$$

Thus

$$(a, b) \in \ker d_1$$

if and only if

$$a - b = 0.$$

Hence

$$a = b.$$

Therefore

$$\ker d_1 = \{(n, n) : n \in \mathbb{Z}\}.$$

But

$$\operatorname{im} d_2 = \{(n, n) : n \in \mathbb{Z}\}.$$

So

$$\ker d_1 = \operatorname{im} d_2.$$

Therefore

$$H_1(C) = 0.$$

At degree 0,

$$H_0(C) = \frac{\mathbb{Z}}{\operatorname{im} d_1}.$$

The map d_1 is surjective because

$$d_1(k, 0) = k.$$

Thus

$$\operatorname{im} d_1 = \mathbb{Z}.$$

Therefore

$$H_0(C) = 0.$$

So the complex is acyclic.

Now we describe the splitting. Inside

$$C_1 = \mathbb{Z}^2,$$

the boundary group is

$$B_1 = \operatorname{im} d_2 = \langle (1, 1) \rangle.$$

Choose the complement

$$A_1 = \langle (1, 0) \rangle.$$

Then

$$C_1 = B_1 \oplus A_1.$$

Indeed, every element $(a, b) \in \mathbb{Z}^2$ can be written as

$$(a, b) = b(1, 1) + (a - b)(1, 0).$$

The restriction

$$d_1|_{A_1}: A_1 \rightarrow C_0$$

is an isomorphism because

$$d_1(1, 0) = 1.$$

Thus the middle group $\mathbb{Z}^2$ splits into one part coming from the previous boundary and one part mapping isomorphically onto the next group.

This is exactly the algebraic mechanism behind the chain contraction.

Example 8.4: A larger exact complex

Consider

$$0 \longrightarrow \mathbb{Z}^2 \xrightarrow{d_2} \mathbb{Z}^3 \xrightarrow{d_1} \mathbb{Z} \longrightarrow 0,$$

where

$$d_2(x, y) = (x, y, -x - y)$$

and

$$d_1(a, b, c) = a + b + c.$$

First check that it is a chain complex:

$$d_1 d_2(x, y) = d_1(x, y, -x - y) = x + y - x - y = 0.$$

Thus

$$d_1 d_2 = 0.$$

Now compute the homology.

At degree 2,

$$H_2(C) = \ker d_2.$$

If

$$d_2(x, y) = 0,$$

then

$$(x, y, -x - y) = (0, 0, 0).$$

Therefore

$$x = 0, \quad y = 0.$$

Hence

$$H_2(C) = 0.$$

At degree 1,

$$H_1(C) = \frac{\ker d_1}{\operatorname{im} d_2}.$$

Now

$$\ker d_1 = \{(a, b, c) \in \mathbb{Z}^3 : a + b + c = 0\}.$$

Every element of this kernel has the form

$$(a, b, -a - b).$$

But

$$(a, b, -a - b) = d_2(a, b).$$

Therefore

$$\ker d_1 = \operatorname{im} d_2.$$

Hence

$$H_1(C) = 0.$$

At degree 0,

$$H_0(C) = \frac{\mathbb{Z}}{\operatorname{im} d_1}.$$

The map d_1 is surjective because

$$d_1(k, 0, 0) = k.$$

Therefore

$$\operatorname{im} d_1 = \mathbb{Z}.$$

Hence

$$H_0(C) = 0.$$

Thus the complex is acyclic and, by the theorem, chain-contractible.

Example 8.5: The augmented chain complex of an interval

Let

$$I = [v_0, v_1]$$

be a closed interval.

The simplicial chain groups are

$$C_1(I) = \mathbb{Z}[e],$$

where

$$e = [v_0, v_1],$$

and

$$C_0(I) = \mathbb{Z}[v_0] \oplus \mathbb{Z}[v_1].$$

The boundary map is

$$\partial_1(e) = v_1 - v_0.$$

The ordinary simplicial chain complex is

$$0 \longrightarrow C_1(I) \xrightarrow{\partial_1} C_0(I) \longrightarrow 0.$$

This complex is not acyclic because the interval is connected, so

$$H_0(I) \cong \mathbb{Z}.$$

To obtain an acyclic complex, we use the augmented chain complex:

$$0 \longrightarrow C_1(I) \xrightarrow{\partial_1} C_0(I) \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0,$$

where the augmentation map is defined by

$$\varepsilon(v_0) = 1, \quad \varepsilon(v_1) = 1.$$

Now check exactness.

First,

$$\varepsilon \partial_1(e) = \varepsilon(v_1 - v_0) = 1 - 1 = 0.$$

Therefore

$$\text{im } \partial_1 \subseteq \ker \varepsilon.$$

Next,

$$\ker \varepsilon = \{av_0 + bv_1 : a + b = 0\}.$$

If $a + b = 0$, then $b = -a$, so

$$av_0 + bv_1 = av_0 - av_1 = -a(v_1 - v_0).$$

But

$$v_1 - v_0 = \partial_1(e).$$

Hence

$$\ker \varepsilon = \text{im } \partial_1.$$

Also,

$$\ker \partial_1 = 0,$$

because

$$\partial_1(e) = v_1 - v_0 \neq 0.$$

Therefore the augmented chain complex of the interval is acyclic.

This is the geometric version of the fact that an interval is contractible.

Example 8.6: The augmented chain complex of a filled triangle

Let

$$\Delta^2 = [v_0, v_1, v_2]$$

be a filled triangle.

The chain groups are

$$C_2(\Delta^2) = \mathbb{Z}[v_0, v_1, v_2],$$

$$C_1(\Delta^2) = \mathbb{Z}[v_0, v_1] \oplus \mathbb{Z}[v_1, v_2] \oplus \mathbb{Z}[v_0, v_2],$$

and

$$C_0(\Delta^2) = \mathbb{Z}[v_0] \oplus \mathbb{Z}[v_1] \oplus \mathbb{Z}[v_2].$$

The boundary of the two-simplex is

$$\partial_2[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1].$$

The boundary of an edge is

$$\partial_1[v_i, v_j] = [v_j] - [v_i].$$

The ordinary homology of the filled triangle is

$$H_0(\Delta^2) \cong \mathbb{Z}, \quad H_1(\Delta^2) = 0, \quad H_2(\Delta^2) = 0.$$

Thus the ordinary chain complex is not acyclic because $H_0 \neq 0$.

However, the augmented chain complex

$$0 \longrightarrow C_2(\Delta^2) \xrightarrow{\partial_2} C_1(\Delta^2) \xrightarrow{\partial_1} C_0(\Delta^2) \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

is acyclic, where

$$\varepsilon(v_0) = \varepsilon(v_1) = \varepsilon(v_2) = 1.$$

Geometrically, the filled triangle has no holes. Its only ordinary homology is the connectedness class in degree 0. The augmentation removes this final H_0 , making the whole augmented complex acyclic.

Therefore the augmented chain complex of Δ^2 is chain-contractible.

Example 8.7: A non-example coming from the circle

Let S^1 be a triangulated circle with vertices

$$v_0, v_1, v_2$$

and edges

$$e_0 = [v_0, v_1], \quad e_1 = [v_1, v_2], \quad e_2 = [v_2, v_0].$$

Then

$$C_1(S^1) = \mathbb{Z}[e_0] \oplus \mathbb{Z}[e_1] \oplus \mathbb{Z}[e_2],$$

and

$$C_0(S^1) = \mathbb{Z}[v_0] \oplus \mathbb{Z}[v_1] \oplus \mathbb{Z}[v_2].$$

The boundary map is

$$\partial_1(e_0) = v_1 - v_0,$$

$$\partial_1(e_1) = v_2 - v_1,$$

$$\partial_1(e_2) = v_0 - v_2.$$

The cycle

$$e_0 + e_1 + e_2$$

has boundary

$$\partial_1(e_0 + e_1 + e_2) = (v_1 - v_0) + (v_2 - v_1) + (v_0 - v_2) = 0.$$

So

$$e_0 + e_1 + e_2 \in \ker \partial_1.$$

But there are no 2-simplices in this triangulation of S^1 , so

$$\operatorname{im} \partial_2 = 0.$$

Therefore

$$H_1(S^1) \cong \mathbb{Z}.$$

The circle has a genuine one-dimensional hole, so its chain complex is not acyclic and cannot be chain-contractible.

This example shows why the condition

$$H_*(C_\bullet) = 0$$

is essential.

2.2 Examples for Problem 9: Products of Polyhedra and Triangulations

Recall the statement:

If

$$K \subset \mathbb{R}^m$$

and

$$L \subset \mathbb{R}^n$$

are simplicial complexes, then

$$|K| \times |L| \subset \mathbb{R}^{m+n}$$

is the polyhedron of a simplicial complex.

The main idea is that

$$|K| \times |L| = \bigcup_{\sigma \in K, \tau \in L} \sigma \times \tau.$$

Each product

$$\sigma \times \tau$$

is a product of simplices, and products of simplices admit staircase triangulations.

Example 9.1: Point times point

Let

$$K = \{[v]\}, \quad L = \{[w]\}.$$

Then

$$|K| = \{v\}, \quad |L| = \{w\}.$$

Therefore

$$|K| \times |L| = \{(v, w)\}.$$

This is a single point in

$$\mathbb{R}^m \times \mathbb{R}^n.$$

Thus

$$\Delta^0 \times \Delta^0 = \Delta^0.$$

So the product is the polyhedron of a simplicial complex consisting of one vertex.

Example 9.2: Point times interval

Let

$$K = \Delta^0 = [v],$$

and

$$L = \Delta^1 = [w_0, w_1].$$

Then

$$|K| \times |L| = \{v\} \times [w_0, w_1].$$

This is an interval with endpoints

$$(v, w_0)$$

and

$$(v, w_1).$$

Thus

$$\Delta^0 \times \Delta^1 \cong \Delta^1.$$

Multiplying by a point does not change the shape.

Example 9.3: Interval times interval gives a square

Let

$$K = \Delta^1 = [v_0, v_1],$$

and

$$L = \Delta^1 = [w_0, w_1].$$

Then

$$|K| \times |L| = [v_0, v_1] \times [w_0, w_1].$$

This is a square.

The four vertices are

$$(v_0, w_0), \quad (v_1, w_0), \quad (v_0, w_1), \quad (v_1, w_1).$$

A square is not a simplex, because a 2-simplex is a triangle. But the square can be triangulated into two triangles.

The two monotone paths from $(0, 0)$ to $(1, 1)$ are

$$(0, 0) \rightarrow (1, 0) \rightarrow (1, 1),$$

and

$$(0, 0) \rightarrow (0, 1) \rightarrow (1, 1).$$

These give the two triangles

$$[(v_0, w_0), (v_1, w_0), (v_1, w_1)]$$

and

$$[(v_0, w_0), (v_0, w_1), (v_1, w_1)].$$

Therefore

$$\Delta^1 \times \Delta^1$$

is the polyhedron of a simplicial complex with two 2-simplices.

Example 9.4: Interval times triangle gives a triangular prism

Let

$$K = \Delta^1 = [v_0, v_1],$$

and

$$L = \Delta^2 = [w_0, w_1, w_2].$$

Then

$$\Delta^1 \times \Delta^2$$

is a triangular prism.

It has vertices

$$(v_0, w_0), \quad (v_0, w_1), \quad (v_0, w_2),$$

and

$$(v_1, w_0), \quad (v_1, w_1), \quad (v_1, w_2).$$

The triangular prism is not a simplex. A 3-simplex is a tetrahedron, but the triangular prism has six vertices.

However, the staircase triangulation divides it into

$$\binom{1+2}{1} = 3$$

tetrahedra.

Thus

$$\Delta^1 \times \Delta^2$$

is a union of three 3-simplices.

Geometrically,

$$\text{interval} \times \text{triangle} = \text{triangular prism}.$$

After triangulation,

$$\text{triangular prism} = \text{three tetrahedra}.$$

Example 9.5: Triangle times triangle

Let

$$K = \Delta^2 = [v_0, v_1, v_2],$$

and

$$L = \Delta^2 = [w_0, w_1, w_2].$$

Then

$$\Delta^2 \times \Delta^2$$

has dimension

$$2 + 2 = 4.$$

Thus it is naturally a 4-dimensional polyhedron.

The staircase triangulation has

$$\binom{2+2}{2} = \binom{4}{2} = 6$$

maximal simplices.

Therefore

$$\Delta^2 \times \Delta^2$$

is triangulated into six 4-simplices.

The maximal simplices correspond to monotone paths from

$$(0, 0)$$

to

$$(2, 2).$$

Each path has four steps: two horizontal steps and two vertical steps. Hence there are

$$6$$

ways to order these steps.

For example, the path

$$(0, 0) \rightarrow (1, 0) \rightarrow (2, 0) \rightarrow (2, 1) \rightarrow (2, 2)$$

gives the simplex

$$[(v_0, w_0), (v_1, w_0), (v_2, w_0), (v_2, w_1), (v_2, w_2)].$$

Another path,

$$(0, 0) \rightarrow (0, 1) \rightarrow (1, 1) \rightarrow (1, 2) \rightarrow (2, 2),$$

gives the simplex

$$[(v_0, w_0), (v_0, w_1), (v_1, w_1), (v_1, w_2), (v_2, w_2)].$$

Altogether, there are six such 4-simplices.

Example 9.6: Circle times interval gives a cylinder

Let K be a triangulated circle. For example, take vertices

$$v_0, v_1, v_2$$

and edges

$$[v_0, v_1], \quad [v_1, v_2], \quad [v_2, v_0].$$

Then

$$|K| \cong S^1.$$

Let

$$L = \Delta^1 = [w_0, w_1].$$

Then

$$|K| \times |L| \cong S^1 \times [0, 1].$$

This is a cylinder.

Each edge of the circle times the interval gives a square:

$$[v_i, v_j] \times [w_0, w_1].$$

Each such square is triangulated into two triangles.

Since the circle has three edges, the cylinder is divided into three squares, and each square is divided into two triangles. Hence this construction gives

$$3 \cdot 2 = 6$$

triangles.

Thus the cylinder is the polyhedron of a simplicial complex.

$$S^1 \times I = \text{cylinder}.$$

Example 9.7: Circle times circle gives a torus

Let K and L both be triangulated circles.

Then

$$|K| \cong S^1, \quad |L| \cong S^1.$$

Therefore

$$|K| \times |L| \cong S^1 \times S^1.$$

This space is a torus:

$$T^2 = S^1 \times S^1.$$

Suppose each circle is triangulated with three edges. Then every pair of edges gives a square:

$$e_i \times f_j.$$

There are

$$3 \cdot 3 = 9$$

such squares.

Each square is triangulated into two triangles. Therefore the product triangulation gives

$$9 \cdot 2 = 18$$

triangles.

Thus the torus is obtained as a product polyhedron and can be triangulated by the product construction.

This is one of the most important examples:

$$S^1 \times S^1 = \text{torus}.$$

Example 9.8: A graph times an interval

Let K be a graph, viewed as a 1-dimensional simplicial complex.

For example, suppose K is shaped like the letter Y . It has one central vertex and three edges coming out of it.

Let

$$L = \Delta^1.$$

Then

$$|K| \times |L|$$

is a thickened graph.

Each edge of K becomes a rectangle:

$$\text{edge} \times \text{interval}.$$

Each rectangle is triangulated into two triangles.

At the central vertex, the rectangular strips meet along a vertical edge.

This example shows that the theorem does not only apply to manifolds. It also applies to polyhedra with singular points, such as graphs with branching.

Example 9.9: Tetrahedron times interval

Let

$$K = \Delta^3 = [v_0, v_1, v_2, v_3],$$

and

$$L = \Delta^1 = [w_0, w_1].$$

Then

$$\Delta^3 \times \Delta^1$$

has dimension

$$3 + 1 = 4.$$

It is a 4-dimensional prism over a tetrahedron.

The staircase triangulation divides it into

$$\binom{3+1}{3} = 4$$

four-dimensional simplices.

Thus

$$\Delta^3 \times \Delta^1$$

is a union of four 4-simplices.

This generalizes the lower-dimensional examples:

$$\Delta^1 \times \Delta^1 = \text{square} = 2 \text{ triangles},$$

$$\Delta^2 \times \Delta^1 = \text{triangular prism} = 3 \text{ tetrahedra},$$

and

$$\Delta^3 \times \Delta^1 = \text{tetrahedral prism} = 4 \text{ four-simplices}.$$

Example 9.10: Product of two general simplicial complexes

Let K be a triangulated circle and let L be a triangulated interval.

Then

$$|K| \times |L|$$

is a cylinder.

But now suppose K is a more complicated 1-dimensional complex, for example a graph with several loops and branches, and suppose L is a triangle Δ^2 .

Then

$$|K| \times |\Delta^2|$$

is built from pieces of the form

$$e \times \Delta^2,$$

where e is an edge of K , and

$$v \times \Delta^2,$$

where v is a vertex of K .

Each product

$$e \times \Delta^2$$

is a triangular prism and is triangulated into three tetrahedra.

Each product

$$v \times \Delta^2$$

is just a triangle.

The compatibility of the staircase triangulation ensures that these tetrahedra and triangles glue correctly along common faces.

Therefore the whole product

$$|K| \times |\Delta^2|$$

is the polyhedron of a simplicial complex.

2.3 Summary of Product Examples

The following table summarizes the main simplex product examples:

Product	Shape	Dimension	Number of maximal simplices
$\Delta^0 \times \Delta^0$	point	0	1
$\Delta^0 \times \Delta^1$	interval	1	1
$\Delta^1 \times \Delta^1$	square	2	2
$\Delta^1 \times \Delta^2$	triangular prism	3	3
$\Delta^2 \times \Delta^2$	4-dimensional product	4	6
$\Delta^3 \times \Delta^1$	4-dimensional prism	4	4

In general,

$$\Delta^p \times \Delta^q$$

has dimension

$$p + q$$

and the staircase triangulation decomposes it into

$$\binom{p+q}{p}$$

maximal simplices.

2.4 Main Intuition

For Problem 8, the condition

$$H_*(C_\bullet) = 0$$

means that every cycle is already a boundary. Since the chain groups are free abelian, the exact sequences split. Therefore the complex decomposes into elementary cancellation pieces of the form

$$\mathbb{Z} \xrightarrow{\cong} \mathbb{Z}.$$

This is why the identity chain map becomes chain homotopic to the zero map.

For Problem 9, the product of two simplices is usually not a simplex, but it can always be triangulated. The simplest nontrivial case is

$$\Delta^1 \times \Delta^1,$$

which is a square divided into two triangles.

The general construction is the staircase triangulation:

$$\Delta^p \times \Delta^q$$

is decomposed into

$$\binom{p+q}{p}$$

simplices.

Then, for arbitrary simplicial complexes K and L , one triangulates every product

$$\sigma \times \tau, \quad \sigma \in K, \quad \tau \in L,$$

and the triangulations glue correctly along common faces. Hence

$$|K| \times |L|$$

is again a polyhedron of a simplicial complex.

3 Diagrams and Charts for Problems 8 and 9

3.1 Diagrams for Problem 8: Acyclic Free Chain Complexes

Diagram 8.1: A chain complex

A chain complex is a sequence of groups and boundary maps satisfying

$$d_n d_{n+1} = 0.$$

$$\cdots \longrightarrow C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \xrightarrow{d_{n-1}} C_{n-2} \longrightarrow \cdots$$

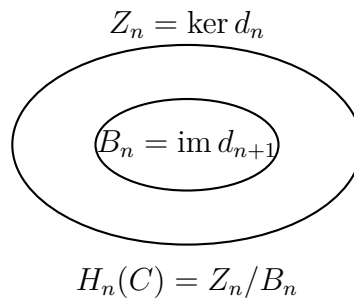
The condition

$$d_n d_{n+1} = 0$$

means that every boundary is a cycle:

$$\text{im } d_{n+1} \subseteq \ker d_n.$$

Diagram 8.2: Cycles, boundaries, and homology



If the complex is acyclic, then

$$H_n(C) = 0.$$

Hence

$$Z_n = B_n.$$

So the smaller region fills the whole cycle group.

Diagram 8.3: Acyclic means exact

If

$$H_*(C_\bullet) = 0,$$

then

$$\ker d_n = \text{im } d_{n+1}$$

for every n . Therefore the complex is exact:

$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & C_{n+1} & \xrightarrow{d_{n+1}} & C_n & \xrightarrow{d_n} & C_{n-1} \xrightarrow{d_{n-1}} \cdots \\
 & & & & \uparrow & & \\
 & & & & \ker d_n = \operatorname{im} d_{n+1} & &
 \end{array}$$

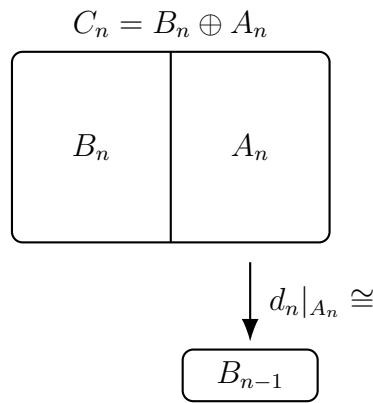
Diagram 8.4: Splitting of C_n

Because each C_n is free abelian and the complex is exact, we have a short exact sequence

$$0 \rightarrow B_n \rightarrow C_n \xrightarrow{d_n} B_{n-1} \rightarrow 0.$$

Since B_{n-1} is free abelian, the sequence splits. Thus

$$C_n \cong B_n \oplus A_n.$$



The important point is that

$$d_n|_{A_n} : A_n \rightarrow B_{n-1}$$

is an isomorphism.

Diagram 8.5: Elementary cancellation pair

Every acyclic free chain complex decomposes into elementary cancellation pieces of the form

$$\mathbb{Z} \xrightarrow{\cong} \mathbb{Z}.$$

Diagrammatically:

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\operatorname{id}} \mathbb{Z} \longrightarrow 0$$

This complex is contractible because the identity map can be undone by a homotopy map going one degree upward.

$$\begin{array}{ccc}
 C_1 = \mathbb{Z} & \xrightarrow{d_1 = \text{id}} & C_0 = \mathbb{Z} \\
 & \nwarrow h_0 & \\
 & &
 \end{array}$$

Here

$$d_1 h_0 = \text{id}_{C_0}, \quad h_0 d_1 = \text{id}_{C_1}.$$

Diagram 8.6: General contraction formula

The chain homotopy condition for

$$\text{id}_{C_\bullet} \simeq 0$$

is

$$\text{id}_{C_n} = d_{n+1} h_n + h_{n-1} d_n.$$

Diagrammatically:

$$\begin{array}{ccccc}
 C_{n+1} & \xrightarrow{d_{n+1}} & C_n & \xrightarrow{d_n} & C_{n-1} \\
 & \nwarrow h_n & & \nwarrow h_{n-1} & \\
 & & & &
 \end{array}$$

The two terms in the formula mean:

$$d_{n+1} h_n$$

recovers the boundary part of an element of C_n , while

$$h_{n-1} d_n$$

recovers the complementary part.

Diagram 8.7: Augmented chain complex of an interval

Let

$$I = [v_0, v_1].$$

The augmented chain complex is

$$0 \rightarrow C_1(I) \xrightarrow{\partial_1} C_0(I) \xrightarrow{\varepsilon} \mathbb{Z} \rightarrow 0.$$

$$0 \longrightarrow \mathbb{Z}[e] \xrightarrow{\partial_1} \mathbb{Z}[v_0] \oplus \mathbb{Z}[v_1] \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

where

$$\partial_1(e) = v_1 - v_0,$$

and

$$\varepsilon(v_0) = 1, \quad \varepsilon(v_1) = 1.$$

Geometrically:

$$\begin{array}{c} \partial e = v_1 - v_0 \\ e = [v_0, v_1] \\ \bullet \text{-----} \bullet \\ v_0 \qquad \qquad v_1 \end{array}$$

The augmentation removes the remaining H_0 , making the augmented complex acyclic.

Chart 8.1: Acyclic and non-acyclic examples

Complex	H_0	H_1	Conclusion
$0 \rightarrow \mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z} \rightarrow 0$	0	0	acyclic and contractible
$0 \rightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \rightarrow 0$	$\mathbb{Z}/2\mathbb{Z}$	0	not acyclic
$0 \rightarrow \mathbb{Z} \xrightarrow{n \mapsto (n,n)} \mathbb{Z}^2 \xrightarrow{(a,b) \mapsto a-b} \mathbb{Z} \rightarrow 0$	0	0	acyclic and contractible

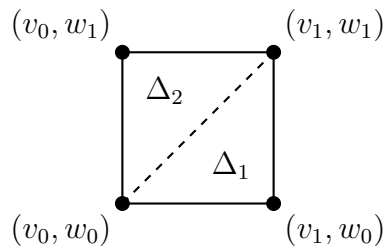
3.2 Diagrams for Problem 9: Products of Polyhedra

Diagram 9.1: Product of two intervals

The product

$$\Delta^1 \times \Delta^1$$

is a square.



The square is triangulated into two triangles:

$$[(v_0, w_0), (v_1, w_0), (v_1, w_1)]$$

and

$$[(v_0, w_0), (v_0, w_1), (v_1, w_1)].$$

Diagram 9.2: Staircase paths for $\Delta^1 \times \Delta^1$

The two triangles correspond to two monotone paths from $(0,0)$ to $(1,1)$.



Thus

$$\Delta^1 \times \Delta^1$$

has

$$\binom{1+1}{1} = 2$$

maximal simplices.

Diagram 9.3: Staircase paths for $\Delta^2 \times \Delta^1$

For

$$\Delta^2 \times \Delta^1,$$

the monotone paths go from

$$(0,0)$$

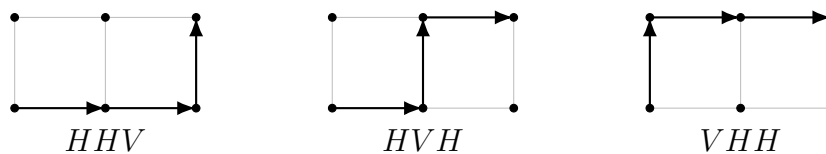
to

$$(2,1).$$

There are

$$\binom{2+1}{2} = 3$$

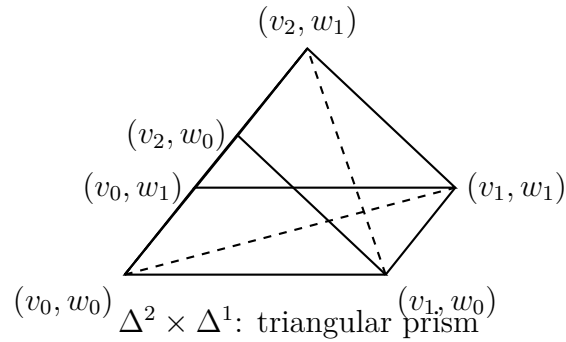
paths.



Therefore the triangular prism

$$\Delta^2 \times \Delta^1$$

is triangulated into three tetrahedra.

Diagram 9.4: Triangular prism from $\Delta^2 \times \Delta^1$ 

The dashed diagonals indicate a possible subdivision into tetrahedra.

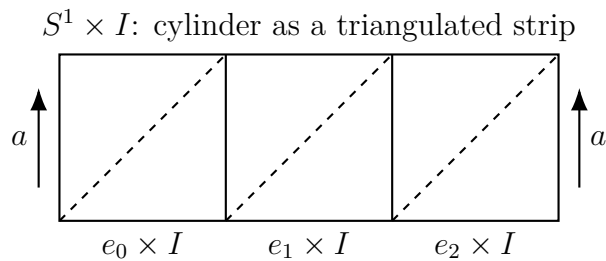
Diagram 9.5: Circle times interval gives a cylinder

Let K be a triangulated circle with three edges. Then

$$|K| \times I$$

is a cylinder.

One may draw it as a rectangle with the left and right edges identified.



There are three rectangles, each divided into two triangles. Hence this triangulation has

$$3 \cdot 2 = 6$$

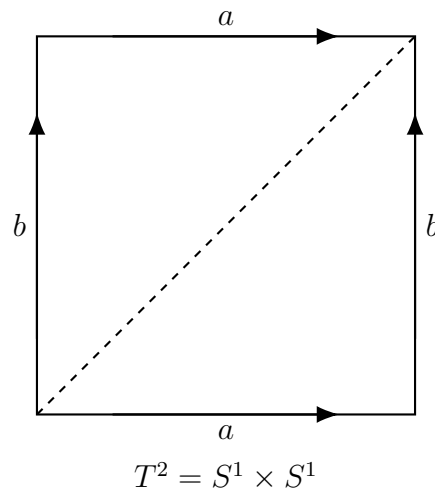
triangles.

Diagram 9.6: Circle times circle gives a torus

The torus

$$T^2 = S^1 \times S^1$$

can be represented by a square whose opposite edges are identified.



This diagram shows the usual quotient-square model of the torus. Product triangulations refine this picture by dividing the square cells into triangles.

Diagram 9.7: Product construction from cells

The product of two polyhedra is built from products of simplices:

$$|K| \times |L| = \bigcup_{\sigma \in K, \tau \in L} \sigma \times \tau.$$

$$\begin{array}{ccccc} \sigma \in K & \xrightarrow{\text{choose}} & \sigma \times \tau & \xrightarrow{\text{triangulate}} & \text{simplices in } K \boxtimes L \\ & \nearrow \text{choose} & & & \\ & \tau \in L & & & \end{array}$$

Each product $\sigma \times \tau$ receives a staircase triangulation.

Chart 9.1: Simplex products

Product	Geometric shape	Dimension	Maximal simplices
$\Delta^0 \times \Delta^0$	point	0	1
$\Delta^0 \times \Delta^1$	interval	1	1
$\Delta^1 \times \Delta^1$	square	2	2
$\Delta^1 \times \Delta^2$	triangular prism	3	3
$\Delta^2 \times \Delta^2$	4-dimensional product	4	6
$\Delta^3 \times \Delta^1$	tetrahedral prism	4	4

In general,

$$\Delta^p \times \Delta^q$$

has dimension

$$p + q$$

and is triangulated into

$$\binom{p+q}{p}$$

maximal simplices.

Chart 9.2: Shape intuition

Product	Intuition
point $\times X$	X
interval $\times$ interval	square
triangle $\times$ interval	triangular prism
$S^1 \times I$	cylinder
$S^1 \times S^1$	torus
graph $\times I$	thickened graph

3.3 Final Visual Summary

Problem	Main idea	Visual model
8	Acyclic free complex	$\mathbb{Z} \xrightarrow{\cong} \mathbb{Z}$ cancellation pairs
9	Product of polyhedra	$\Delta^1 \times \Delta^1 = \text{square} = 2 \text{ triangles}$

Problem 8 is algebraic: exactness plus freeness gives splittings and contractions.

Problem 9 is geometric: products of simplices are triangulated by staircase paths, and these triangulations glue together across faces.

4 Elementary Collapses, Free Faces, and Homology

4.1 Main geometric idea

Let K be a simplicial complex. Suppose that

$$\tau \leq \sigma,$$

where τ is a face of σ of codimension 1. Thus, if

$$\dim \sigma = q,$$

then

$$\dim \tau = q - 1.$$

Assume that σ is a maximal simplex of K , meaning that σ is not a proper face of any larger simplex in K . Also assume that τ is a face of no simplex of K other than σ .

Such a face τ is called a **free face** of σ .

We define

$$L = K \setminus \{\sigma, \tau\}.$$

The operation

$$K \searrow L$$

is called an **elementary collapse**. Geometrically, we remove one maximal simplex together with one of its free codimension-one faces.

The main result is that this operation does not change homology:

$$H_j(L) \cong H_j(K)$$

for every j .

4.2 Why L is still a simplicial complex

We first check that L is a simplicial complex.

Recall that a simplicial complex must contain all faces of its simplices. Therefore, if

$$\rho \in L$$

and

$$\eta \leq \rho,$$

then we must show that

$$\eta \in L.$$

Since $L \subset K$, we know that $\rho \in K$. Because K is a simplicial complex and $\eta \leq \rho$, we have

$$\eta \in K.$$

The only possible problem is that η might be one of the two simplices removed from K , namely σ or τ .

First, $\eta = \sigma$ is impossible. Indeed, $\eta \leq \rho$, while $\rho \neq \sigma$, because $\rho \in L$. A face of ρ cannot be a larger simplex σ .

Second, suppose that

$$\eta = \tau.$$

Then

$$\tau \leq \rho.$$

But τ is assumed to be a face of no simplex of K other than σ . Hence $\rho = \sigma$, which is impossible because $\rho \in L$.

Therefore

$$\eta \neq \sigma, \tau.$$

So

$$\eta \in L.$$

Thus L is again a simplicial complex.

4.3 Chain-level description

Let

$$i: L \hookrightarrow K$$

be the inclusion. It induces a chain map

$$i_{\bullet}: C_{\bullet}(L) \rightarrow C_{\bullet}(K).$$

Since K and L differ only by the two simplices σ and τ , the chain groups split as follows.

In degree q ,

$$C_q(K) = C_q(L) \oplus \mathbb{Z}\sigma.$$

In degree $q - 1$,

$$C_{q-1}(K) = C_{q-1}(L) \oplus \mathbb{Z}\tau.$$

In all other degrees,

$$C_j(K) = C_j(L) \quad \text{for } j \neq q, q - 1.$$

The key boundary relation is

$$\partial\sigma = \pm\tau + a,$$

where

$$a \in C_{q-1}(L).$$

After choosing the orientation of σ suitably, we may assume that the coefficient of τ is $+1$. Hence

$$\partial\sigma = \tau + a.$$

Here a is a chain made from the other codimension-one faces of σ . These other faces belong to L .

Because

$$\partial^2 = 0,$$

we have

$$0 = \partial^2\sigma = \partial(\tau + a).$$

Therefore

$$0 = \partial\tau + \partial a.$$

Hence

$$\partial\tau = -\partial a.$$

This identity is the algebraic reason why the elementary collapse preserves homology.

4.4 Construction of a chain retraction

We now define a chain map

$$r_\bullet : C_\bullet(K) \rightarrow C_\bullet(L)$$

which behaves like a retraction.

For every simplex $x \in L$, define

$$r(x) = x.$$

For the removed q -simplex σ , define

$$r(\sigma) = 0.$$

For the removed free face τ , define

$$r(\tau) = -a,$$

where

$$\partial\sigma = \tau + a.$$

Thus r fixes everything in L , kills σ , and replaces the free face τ by the negative of the remaining boundary of σ .

We check that r is a chain map.

First, for σ , we have

$$r(\partial\sigma) = r(\tau + a) = r(\tau) + r(a).$$

Since $a \in C_{q-1}(L)$, we have $r(a) = a$. Also $r(\tau) = -a$. Therefore

$$r(\partial\sigma) = -a + a = 0.$$

On the other hand,

$$\partial r(\sigma) = \partial 0 = 0.$$

Hence

$$r\partial\sigma = \partial r\sigma.$$

Next, for τ , we have

$$r(\partial\tau) = \partial\tau,$$

because $\partial\tau \in C_{q-2}(L)$. Also,

$$\partial r(\tau) = \partial(-a) = -\partial a.$$

But from

$$\partial\tau = -\partial a,$$

we get

$$\partial r(\tau) = \partial\tau.$$

Therefore

$$r\partial\tau = \partial r\tau.$$

For simplices already in L , the equality

$$r\partial = \partial r$$

is immediate, because r is the identity on $C_\bullet(L)$.

Thus $r_\bullet$ is a chain map.

Moreover,

$$r_\bullet i_\bullet = \text{id}_{C_\bullet(L)}.$$

So $r_\bullet$ is a left inverse of the inclusion on the chain level.

4.5 Chain homotopy between ir and the identity

We now show that

$$i_\bullet r_\bullet$$

is chain homotopic to

$$\text{id}_{C_\bullet(K)}.$$

Define homomorphisms

$$h_j: C_j(K) \rightarrow C_{j+1}(K)$$

by

$$h(\tau) = \sigma$$

and

$$h(x) = 0$$

for every other basis simplex x .

We claim that

$$\text{id}_{C_\bullet(K)} - i_\bullet r_\bullet = \partial h + h \partial.$$

We check this on basis simplices.

First, consider σ . Since

$$r(\sigma) = 0,$$

we get

$$(\text{id} - ir)(\sigma) = \sigma.$$

On the other hand,

$$(\partial h + h \partial)(\sigma) = \partial h(\sigma) + h(\partial \sigma).$$

Because

$$h(\sigma) = 0$$

and

$$\partial \sigma = \tau + a,$$

we have

$$h(\partial \sigma) = h(\tau + a) = h(\tau) + h(a).$$

Since $a \in C_{q-1}(L)$, we have $h(a) = 0$. Therefore

$$h(\partial\sigma) = \sigma.$$

Thus

$$(\partial h + h\partial)(\sigma) = \sigma.$$

So the formula holds on σ .

Now consider τ . Since

$$r(\tau) = -a,$$

we get

$$(\text{id} - ir)(\tau) = \tau - ir(\tau) = \tau - (-a) = \tau + a.$$

On the other hand,

$$(\partial h + h\partial)(\tau) = \partial h(\tau) + h(\partial\tau).$$

Since

$$h(\tau) = \sigma,$$

we get

$$\partial h(\tau) = \partial\sigma = \tau + a.$$

Also, $\partial\tau \in C_{q-2}(L)$, so

$$h(\partial\tau) = 0.$$

Therefore

$$(\partial h + h\partial)(\tau) = \tau + a.$$

So the formula holds on τ .

Finally, let $x \in L$. Since $r(x) = x$, we have

$$(\text{id} - ir)(x) = x - x = 0.$$

Also, $h(x) = 0$. Since L is a subcomplex, $\partial x \in C_\bullet(L)$, so

$$h(\partial x) = 0.$$

Thus

$$(\partial h + h\partial)(x) = 0.$$

Therefore

$$\text{id}_{C_\bullet(K)} - i_\bullet r_\bullet = \partial h + h\partial.$$

Hence

$$i_\bullet r_\bullet \simeq \text{id}_{C_\bullet(K)}.$$

Together with

$$r_\bullet i_\bullet = \text{id}_{C_\bullet(L)},$$

this shows that

$$C_\bullet(K)$$

and

$$C_{\bullet}(L)$$

are chain homotopy equivalent.

Therefore

$$H_j(K) \cong H_j(L)$$

for every j .

4.6 Relative homology viewpoint

There is also a shorter proof using relative homology.

Consider the relative chain complex

$$C_{\bullet}(K, L) = C_{\bullet}(K)/C_{\bullet}(L).$$

Since K and L differ only by σ and τ , the quotient complex has only two nonzero groups:

$$C_q(K, L) \cong \mathbb{Z}[\sigma],$$

and

$$C_{q-1}(K, L) \cong \mathbb{Z}[\tau].$$

The boundary map is

$$\partial[\sigma] = \pm[\tau].$$

Thus the relative chain complex is of the form

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \longrightarrow 0.$$

This complex is acyclic. Therefore

$$H_j(K, L) = 0$$

for every j .

Now use the long exact sequence of the pair (K, L) :

$$\cdots \longrightarrow H_j(L) \xrightarrow{i_*} H_j(K) \longrightarrow H_j(K, L) \longrightarrow H_{j-1}(L) \longrightarrow \cdots$$

Since

$$H_j(K, L) = 0$$

for all j , the map

$$i_*: H_j(L) \rightarrow H_j(K)$$

is an isomorphism for every j .

Therefore elementary collapse preserves homology.

4.7 Example 1: A filled triangle collapses to two edges

Let

$$K = \Delta^2$$

be a filled triangle.

Choose one boundary edge τ , and let

$$\sigma = \Delta^2.$$

The edge τ is a free face because it belongs only to the triangle σ .

Now remove both:

$$L = K \setminus \{\sigma, \tau\}.$$

What remains is two edges meeting at one vertex. This space is shaped like the letter V , so it is contractible.

Therefore K and L have the same homology:

$$H_0(K) \cong H_0(L) \cong \mathbb{Z},$$

and

$$H_j(K) \cong H_j(L) = 0 \quad \text{for } j \geq 1.$$

The elementary collapse removes no genuine hole.

4.8 Example 2: A tetrahedron collapses by removing one face

Let

$$K = \Delta^3$$

be a filled tetrahedron.

Take

$$\sigma = \Delta^3,$$

and let τ be one triangular face of σ .

The face τ is free because it belongs only to the tetrahedron σ .

Removing σ and τ leaves the other three triangular faces. This remaining complex is still contractible, geometrically like a disk.

Hence

$$H_0(K) \cong \mathbb{Z},$$

and

$$H_j(K) = 0 \quad \text{for } j \geq 1.$$

The same homology groups are obtained for the collapsed complex L .

4.9 Example 3: A triangulated disk

Suppose K is a triangulated disk.

Usually, a triangle at the boundary has a boundary edge that belongs only to that triangle. Such an edge is a free face.

We may remove that triangle together with the free edge. Then we repeat this process.

In many triangulations of a disk, repeated elementary collapses reduce the whole complex to a single vertex:

$$K \searrow \{\text{point}\}.$$

Therefore a triangulated disk has the same homology as a point:

$$H_0(K) \cong \mathbb{Z},$$

and

$$H_j(K) = 0 \quad \text{for } j \geq 1.$$

This matches the geometric intuition that a disk is connected and has no holes.

4.10 Example 4: A circle has no free face

Let K be a triangulated circle.

In a one-dimensional simplicial complex, an elementary collapse removes an edge together with a free vertex.

But in a circle, every vertex belongs to two edges. Therefore no vertex is free.

Hence a triangulated circle has no elementary collapse of this type.

The homology of the circle is

$$H_0(S^1) \cong \mathbb{Z},$$

$$H_1(S^1) \cong \mathbb{Z},$$

and

$$H_j(S^1) = 0 \quad \text{for } j \geq 2.$$

The generator of $H_1(S^1)$ is represented by the cycle going once around the circle.

This example shows that elementary collapses cannot remove genuine holes.

4.11 How collapses help compute homology

Elementary collapses are useful because they simplify a complex without changing its homology.

The general strategy is:

$$\text{complicated complex} \searrow \text{simpler complex} \searrow \text{minimal core}.$$

Since every elementary collapse preserves homology, we may compute the homology of the final simplified complex instead of the original one.

Typical outcomes are as follows.

If a complex collapses to a point, then

$$H_0 \cong \mathbb{Z},$$

and

$$H_j = 0 \quad \text{for } j \geq 1.$$

If a complex collapses to a circle S^1 , then

$$H_0 \cong \mathbb{Z},$$

$$H_1 \cong \mathbb{Z},$$

and

$$H_j = 0 \quad \text{for } j \geq 2.$$

If a complex collapses to a wedge of two circles

$$S^1 \vee S^1,$$

then

$$H_0 \cong \mathbb{Z},$$

$$H_1 \cong \mathbb{Z}^2,$$

and

$$H_j = 0 \quad \text{for } j \geq 2.$$

If a complex collapses to a sphere S^2 , then

$$H_0 \cong \mathbb{Z},$$

$$H_2 \cong \mathbb{Z},$$

and

$$H_j = 0 \quad \text{for } j \neq 0, 2.$$

If a complex collapses to a torus T^2 , then

$$H_0 \cong \mathbb{Z},$$

$$H_1 \cong \mathbb{Z}^2,$$

$$H_2 \cong \mathbb{Z},$$

and

$$H_j = 0 \quad \text{for } j \geq 3.$$

4.12 How to describe homology generators

After simplifying a complex by collapses, the generators of homology are described by the remaining cycles.

For H_0 , a generator is represented by a vertex in each connected component. If the complex is connected, then

$$H_0 \cong \mathbb{Z}$$

with generator represented by any vertex

$$[v].$$

For H_1 , a generator is represented by a loop of edges. For example, in a triangulated circle,

$$e_1 + e_2 + \cdots + e_n$$

represents a generator of

$$H_1(S^1) \cong \mathbb{Z}.$$

For H_2 , a generator is represented by a closed oriented surface. For example, in a triangulated sphere,

$$\sum_i \pm \sigma_i$$

where the sum is taken over all oriented triangular faces, represents a generator of

$$H_2(S^2) \cong \mathbb{Z}.$$

For a torus,

$$H_1(T^2) \cong \mathbb{Z}^2.$$

The two generators are represented by two fundamental loops:

$$a = \text{loop around one direction,}$$

and

$$b = \text{loop around the other direction.}$$

Also,

$$H_2(T^2) \cong \mathbb{Z}$$

is generated by the oriented sum of all triangular faces of the torus.

4.13 Core intuition

An elementary collapse removes a pair

$$(\sigma, \tau),$$

where σ is a simplex and τ is a free codimension-one face of σ .

Algebraically, this pair behaves like a cancellation pair:

$$\mathbb{Z}[\sigma] \xrightarrow{\partial} \mathbb{Z}[\tau].$$

Since

$$\partial\sigma = \pm\tau + \text{terms already in } L,$$

the simplex σ kills the face τ exactly. Removing both does not remove any genuine cycle and does not create any new boundary.

Thus the main slogan is:

Free faces can be collapsed without changing homology.

Elementary collapses are powerful because they reduce complicated simplicial complexes to much simpler ones while preserving all homology groups.

5 Examples of Elementary Collapses

In this section we give simple, medium, and harder examples of elementary collapses and their effect on homology.

Recall the situation:

$$K \searrow L = K \setminus \{\sigma, \tau\},$$

where σ is a maximal simplex of K , and $\tau \leq \sigma$ is a free codimension-one face of σ .

The main theorem says that the inclusion

$$i: L \hookrightarrow K$$

induces isomorphisms

$$i_*: H_j(L) \xrightarrow{\cong} H_j(K)$$

for every j .

Thus elementary collapses preserve homology.

5.1 Simple Examples

Example 1.1: An interval collapses to a point

Let

$$K = [v_0, v_1]$$

be one closed edge.

The simplices of K are

$$[v_0], \quad [v_1], \quad [v_0, v_1].$$

Take

$$\sigma = [v_0, v_1], \quad \tau = [v_1].$$

The vertex $[v_1]$ is a free face because it belongs only to the edge $[v_0, v_1]$.

Remove both:

$$L = K \setminus \{[v_0, v_1], [v_1]\}.$$

Then

$$L = \{[v_0]\}.$$

Hence

$$[v_0, v_1] \searrow [v_0].$$

Before the collapse, K is an interval, so it is contractible. Therefore

$$H_0(K) \cong \mathbb{Z},$$

and

$$H_j(K) = 0 \quad \text{for } j \geq 1.$$

After the collapse, L is a single point, so

$$H_0(L) \cong \mathbb{Z},$$

and

$$H_j(L) = 0 \quad \text{for } j \geq 1.$$

Thus the collapse preserves homology.

Example 1.2: A filled triangle collapses to two edges

Let

$$K = \Delta^2 = [v_0, v_1, v_2]$$

be a filled triangle.

Choose

$$\sigma = [v_0, v_1, v_2], \quad \tau = [v_1, v_2].$$

The edge $[v_1, v_2]$ belongs only to the triangle $[v_0, v_1, v_2]$, so it is a free face.

Remove both:

$$L = K \setminus \{[v_0, v_1, v_2], [v_1, v_2]\}.$$

What remains is

$$[v_0, v_1] \cup [v_0, v_2].$$

Geometrically, this is shaped like the letter V . Hence

$$\Delta^2 \searrow [v_0, v_1] \cup [v_0, v_2].$$

The filled triangle and the V -shape are both contractible. Therefore

$$H_0(K) \cong H_0(L) \cong \mathbb{Z},$$

and

$$H_j(K) \cong H_j(L) = 0 \quad \text{for } j \geq 1.$$

Example 1.3: The V-shape collapses to a point

Now take the remaining complex

$$L = [v_0, v_1] \cup [v_0, v_2].$$

Choose

$$\sigma = [v_0, v_1], \quad \tau = [v_1].$$

The vertex $[v_1]$ is free because it belongs only to the edge $[v_0, v_1]$.

Remove both:

$$L_1 = L \setminus \{[v_0, v_1], [v_1]\}.$$

Then

$$L_1 = [v_0, v_2].$$

Now choose

$$\sigma = [v_0, v_2], \quad \tau = [v_2].$$

The vertex $[v_2]$ is a free face of the edge $[v_0, v_2]$. Remove both:

$$L_2 = L_1 \setminus \{[v_0, v_2], [v_2]\}.$$

Then

$$L_2 = \{[v_0]\}.$$

Thus the filled triangle collapses all the way to a point:

$$\Delta^2 \searrow [v_0, v_1] \cup [v_0, v_2] \searrow [v_0, v_2] \searrow [v_0].$$

Therefore

$$\Delta^2 \searrow \{\text{point}\}.$$

Hence

$$H_0(\Delta^2) \cong \mathbb{Z},$$

and

$$H_j(\Delta^2) = 0 \quad \text{for } j \geq 1.$$

5.2 Medium Examples**Example 2.1: A square divided into two triangles collapses to a point**

Let K be a square triangulated by one diagonal.

Let the vertices be

$$v_0, v_1, v_2, v_3,$$

and suppose the square is divided into the two triangles

$$[v_0, v_1, v_2]$$

and

$$[v_0, v_2, v_3].$$

Thus

$$K = [v_0, v_1, v_2] \cup [v_0, v_2, v_3].$$

The diagonal is

$$[v_0, v_2].$$

Since K is a filled square, it should collapse to a point.

First choose

$$\sigma_1 = [v_0, v_1, v_2], \quad \tau_1 = [v_1, v_2].$$

The edge $[v_1, v_2]$ is a boundary edge and belongs only to σ_1 , so it is a free face.

Remove both:

$$K_1 = K \setminus \{\sigma_1, \tau_1\}.$$

Now choose

$$\sigma_2 = [v_0, v_2, v_3], \quad \tau_2 = [v_2, v_3].$$

The edge $[v_2, v_3]$ is now free.

Remove both:

$$K_2 = K_1 \setminus \{\sigma_2, \tau_2\}.$$

What remains is a tree of edges, for example

$$[v_0, v_1] \cup [v_0, v_2] \cup [v_0, v_3].$$

A tree collapses to a point by repeatedly removing a leaf edge and its leaf vertex.

Therefore

$$K \searrow \{\text{point}\}.$$

Hence

$$H_0(K) \cong \mathbb{Z},$$

and

$$H_j(K) = 0 \quad \text{for } j \geq 1.$$

Example 2.2: A triangulated disk collapses to a point

Let K be a triangulated disk.

For example, imagine a polygon cut into many triangles. A boundary triangle usually has at least one boundary edge that belongs only to that triangle. Such an edge is a free face.

Thus we can remove a pair of the form

$$(\text{boundary triangle, free boundary edge}).$$

After removing one such pair, we obtain a smaller triangulated disk or a smaller contractible complex. We then repeat the process.

Each elementary collapse removes one triangle and one free edge.

Eventually, a collapsible triangulated disk reduces to a graph. Since a disk has no essential loop, the remaining graph can be reduced to a tree, and the tree collapses to a point.

Thus, for a collapsible triangulated disk,

$$K \searrow \{\text{point}\}.$$

Therefore

$$H_0(K) \cong \mathbb{Z},$$

and

$$H_j(K) = 0 \quad \text{for } j \geq 1.$$

This is a central example: disks are homologically trivial except in degree 0.

Example 2.3: A tree collapses to a point

Let K be a finite tree.

A tree is a connected graph with no cycles.

Every finite tree has at least one leaf vertex. A leaf vertex is a vertex that belongs to exactly one edge.

Let v be a leaf vertex, and let

$$e = [v, w]$$

be the unique edge containing v .

Then

$$\tau = [v]$$

is a free face of

$$\sigma = e.$$

Remove both:

$$K_1 = K \setminus \{e, v\}.$$

The result is again a tree, unless the original tree was just one edge.

Repeating this process, every finite tree collapses to a point:

$$K \searrow \{\text{point}\}.$$

Therefore

$$H_0(K) \cong \mathbb{Z},$$

and

$$H_1(K) = 0.$$

This example is important because it shows that collapses detect the absence of loops in a graph.

Example 2.4: A circle does not collapse to a point

Let K be a triangulated circle with vertices

$$v_0, v_1, v_2$$

and edges

$$[v_0, v_1], \quad [v_1, v_2], \quad [v_2, v_0].$$

In a graph, an elementary collapse removes an edge together with a free vertex.

But every vertex of a circle belongs to two edges. Therefore there is no free vertex.

Hence a triangulated circle has no elementary collapse.

Its homology is

$$H_0(S^1) \cong \mathbb{Z},$$

$$H_1(S^1) \cong \mathbb{Z},$$

and

$$H_j(S^1) = 0 \quad \text{for } j \geq 2.$$

The nonzero group $H_1(S^1)$ corresponds to the loop going once around the circle.

This example shows that genuine holes cannot be removed by elementary collapses.

5.3 Harder Examples**Example 3.1: An annulus collapses to a circle**

Let K be a triangulated annulus.

Geometrically, an annulus is a cylinder-like band:

$$S^1 \times [0, 1].$$

It has one essential circular hole.

By elementary collapses, one can remove triangles from the boundary of the annulus, peeling off layers, until the annulus collapses to its central circle:

$$K \searrow S^1.$$

Therefore

$$H_0(K) \cong \mathbb{Z},$$

$$H_1(K) \cong \mathbb{Z},$$

and

$$H_j(K) = 0 \quad \text{for } j \geq 2.$$

The collapse cannot remove the central loop, because that loop represents a genuine H_1 -class.

Thus the annulus is not collapsible to a point, but it collapses to a circle.

Example 3.2: A solid tetrahedron collapses to a point

Let

$$K = \Delta^3 = [v_0, v_1, v_2, v_3]$$

be a filled tetrahedron.

Choose one triangular face, for example

$$\tau = [v_1, v_2, v_3].$$

This is a free face of the tetrahedron

$$\sigma = [v_0, v_1, v_2, v_3].$$

Remove both:

$$K_1 = K \setminus \{\sigma, \tau\}.$$

Now K_1 consists of the other three triangular faces:

$$[v_0, v_1, v_2], \quad [v_0, v_1, v_3], \quad [v_0, v_2, v_3].$$

This remaining complex is like a triangular disk made of three triangles. It collapses further to a tree and then to a point.

Therefore

$$\Delta^3 \searrow \{\text{point}\}.$$

So

$$H_0(\Delta^3) \cong \mathbb{Z},$$

and

$$H_j(\Delta^3) = 0 \quad \text{for } j \geq 1.$$

This is the three-dimensional analogue of the filled triangle.

Example 3.3: The boundary of a tetrahedron does not collapse to a point

Now let K be not the filled tetrahedron, but only its boundary:

$$K = \partial\Delta^3.$$

This is a triangulated sphere:

$$\partial\Delta^3 \cong S^2.$$

It consists of four triangular faces:

$$[v_1, v_2, v_3],$$

$$[v_0, v_2, v_3],$$

$$[v_0, v_1, v_3],$$

and

$$[v_0, v_1, v_2].$$

Every edge belongs to exactly two triangles.

Therefore no edge is a free face of a triangle. Hence there is no elementary collapse of dimension 2.

This reflects the fact that S^2 has genuine 2-dimensional homology:

$$H_0(S^2) \cong \mathbb{Z},$$

$$H_2(S^2) \cong \mathbb{Z},$$

and

$$H_1(S^2) = 0.$$

The group $H_2(S^2)$ is generated by the oriented sum of all triangular faces.

So the boundary of a tetrahedron cannot collapse to a point because it contains a closed surface cycle.

Example 3.4: A torus cannot collapse to a point

Let K be a triangulated torus:

$$K \cong T^2.$$

The torus has homology

$$H_0(T^2) \cong \mathbb{Z},$$

$$H_1(T^2) \cong \mathbb{Z}^2,$$

$$H_2(T^2) \cong \mathbb{Z}.$$

The group $H_1(T^2)$ has two independent generators:

$$a = \text{meridian loop},$$

and

$$b = \text{longitude loop}.$$

The group $H_2(T^2)$ is generated by the oriented sum of all triangular faces.

If the torus collapsed to a point, then it would have the same homology as a point:

$$H_0 \cong \mathbb{Z},$$

and

$$H_j = 0 \quad \text{for } j \geq 1.$$

But the torus has nonzero H_1 and nonzero H_2 . Therefore a triangulated torus cannot collapse to a point.

This example shows how homology obstructs collapsibility.

Example 3.5: A cone collapses to its apex

Let A be a simplicial complex, and let

$$K = \text{Cone}(A)$$

be the cone on A .

This means we add a new vertex p , called the apex, and join p to every simplex of A .

For example:

$$\text{Cone}(S^1)$$

is a disk, and

$$\text{Cone}(S^2)$$

is a ball.

Cones are always contractible. In many finite simplicial cases, a cone can be collapsed to its apex:

$$\text{Cone}(A) \searrow \{p\}.$$

Therefore

$$H_0(\text{Cone}(A)) \cong \mathbb{Z},$$

and

$$H_j(\text{Cone}(A)) = 0 \quad \text{for } j \geq 1.$$

This is a powerful example because it explains why attaching a cone kills homology.

For instance, S^1 has a nontrivial loop:

$$H_1(S^1) \cong \mathbb{Z}.$$

But the cone over S^1 fills this loop with a disk. Therefore

$$H_1(\text{Cone}(S^1)) = 0.$$

5.4 Hard Algebraic Example: Collapse as Chain Cancellation

Let K be a simplicial complex with a simplex σ and a free face τ , where

$$\dim \sigma = q, \quad \dim \tau = q - 1.$$

Let

$$L = K \setminus \{\sigma, \tau\}.$$

Suppose that, after choosing orientations,

$$\partial \sigma = \tau + a,$$

where

$$a \in C_{q-1}(L).$$

Then the relative chain complex

$$C_{\bullet}(K, L)$$

has only two nonzero groups:

$$C_q(K, L) = \mathbb{Z}[\sigma],$$

and

$$C_{q-1}(K, L) = \mathbb{Z}[\tau].$$

The boundary map is

$$\partial[\sigma] = [\tau].$$

Thus $C_{\bullet}(K, L)$ has the form

$$0 \longrightarrow \mathbb{Z}[\sigma] \xrightarrow{\cong} \mathbb{Z}[\tau] \longrightarrow 0.$$

This complex is acyclic, so

$$H_j(K, L) = 0 \quad \text{for all } j.$$

Now consider the long exact sequence of the pair (K, L) :

$$\cdots \longrightarrow H_j(L) \xrightarrow{i_*} H_j(K) \longrightarrow H_j(K, L) \longrightarrow H_{j-1}(L) \longrightarrow \cdots.$$

Since

$$H_j(K, L) = 0$$

for all j , it follows that

$$i_*: H_j(L) \rightarrow H_j(K)$$

is an isomorphism for every j .

Thus an elementary collapse is exactly the removal of an acyclic two-term summand:

$$\mathbb{Z}[\sigma] \xrightarrow{\cong} \mathbb{Z}[\tau].$$

5.5 Summary Chart

Complex	Collapse result	Homology	Difficulty
interval	point	$H_0 \cong \mathbb{Z}$	simple
filled triangle	point	$H_0 \cong \mathbb{Z}$	simple
tree	point	$H_0 \cong \mathbb{Z}$	medium
circle	no collapse to point	$H_0 \cong \mathbb{Z}, H_1 \cong \mathbb{Z}$	medium
annulus	circle	$H_0 \cong \mathbb{Z}, H_1 \cong \mathbb{Z}$	harder
filled tetrahedron	point	$H_0 \cong \mathbb{Z}$	harder
$\partial\Delta^3 \cong S^2$	no collapse to point	$H_0 \cong \mathbb{Z}, H_2 \cong \mathbb{Z}$	harder
T^2	no collapse to point	$H_0 \cong \mathbb{Z}, H_1 \cong \mathbb{Z}^2, H_2 \cong \mathbb{Z}$	hard

5.6 Key Intuition

An elementary collapse removes a pair

$$(\sigma, \tau),$$

where σ is a simplex and τ is a free codimension-one face of σ .

This pair contributes only an acyclic chain piece:

$$\mathbb{Z}[\sigma] \xrightarrow{\cong} \mathbb{Z}[\tau].$$

Therefore removing this pair does not change homology.

However, genuine loops and closed surfaces cannot be removed this way. This is why

a disk collapses to a point,

but

$$S^1$$

does not collapse to a point, and

$$S^2$$

does not collapse to a point.

The main rule is:

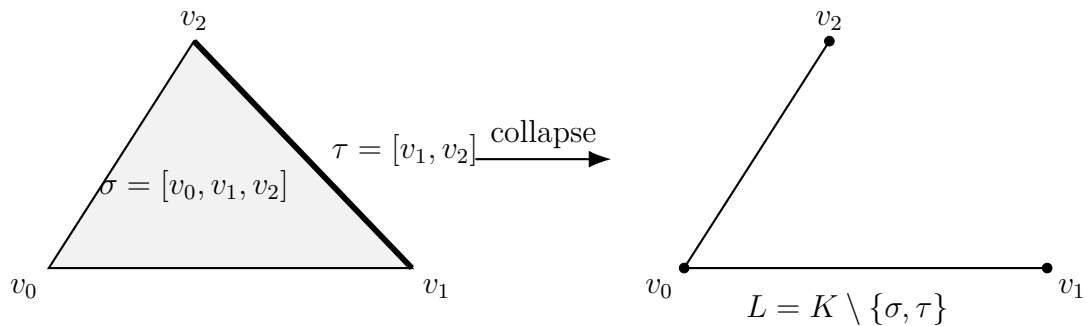
Collapses remove contractible pieces, not genuine holes.

6 Diagrams and Charts for Elementary Collapses

6.1 Diagram 1: Free face and elementary collapse

An elementary collapse removes a maximal simplex σ together with a free codimension-one face τ :

$$K \searrow L = K \setminus \{\sigma, \tau\}.$$



The edge τ is free because it belongs only to the triangle σ . Removing both does not change homology.

6.2 Diagram 2: Interval collapses to a point

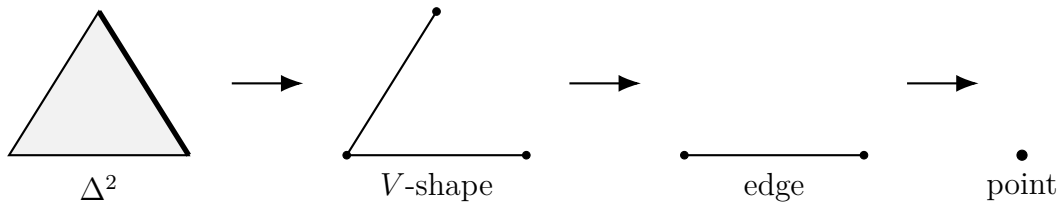
$$[v_0, v_1] \searrow [v_0].$$

$$\begin{array}{ccc} v_0 & \xrightarrow{\sigma = [v_0, v_1]} v_1 & \xrightarrow{\tau = [v_0] \text{ collapse}} v_0 \\ H_0 \cong \mathbb{Z}, & H_j = 0 \text{ for } j \geq 1 & \end{array}$$

6.3 Diagram 3: Filled triangle collapses to a point

A filled triangle collapses by removing a triangle with a free edge, then removing leaf edges.

$$\Delta^2 \searrow V\text{-shape} \searrow \text{edge} \searrow \text{point}.$$



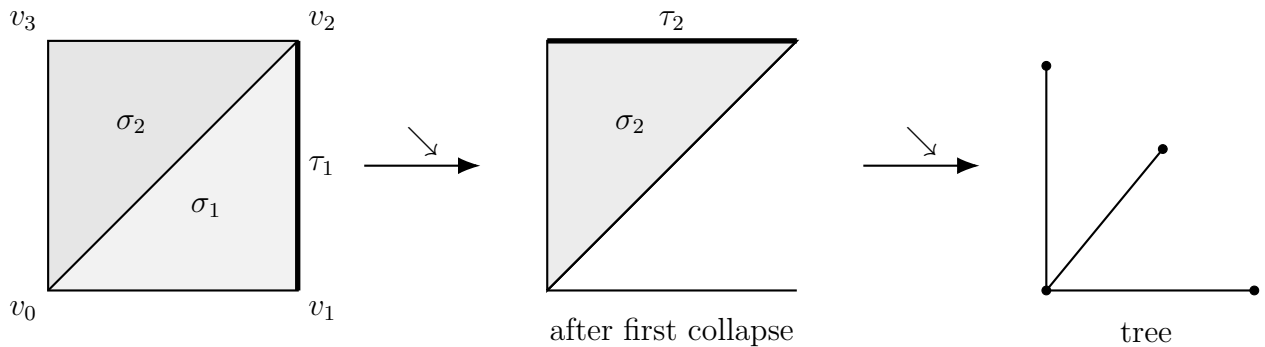
Thus

$$H_0(\Delta^2) \cong \mathbb{Z}, \quad H_j(\Delta^2) = 0 \quad j \geq 1.$$

6.4 Diagram 4: Square triangulated by a diagonal collapses to a point

Let the square be triangulated by the diagonal $[v_0, v_2]$:

$$K = [v_0, v_1, v_2] \cup [v_0, v_2, v_3].$$

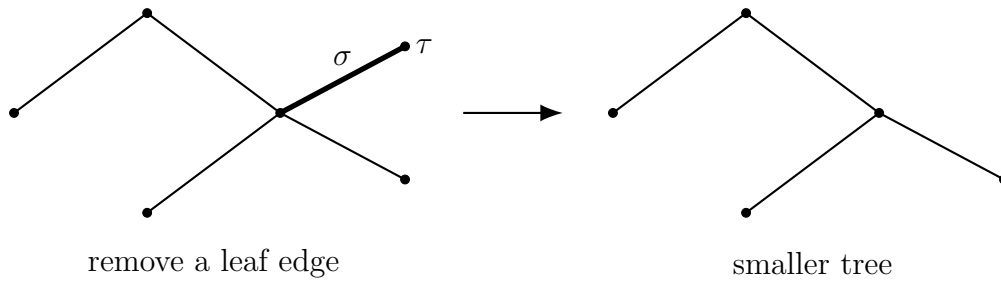


The remaining tree collapses to a point, so the whole square has the homology of a point.

$$H_0(K) \cong \mathbb{Z}, \quad H_j(K) = 0 \quad j \geq 1.$$

6.5 Diagram 5: Tree collapses by deleting leaf edges

A tree collapses by repeatedly removing a leaf edge and its leaf vertex.

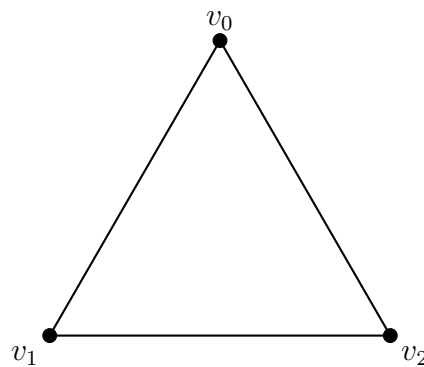


Repeating this process gives

$$K \searrow \{\text{point}\}.$$

6.6 Diagram 6: Circle has no free vertex

A triangulated circle has no free vertex, because every vertex belongs to two edges.



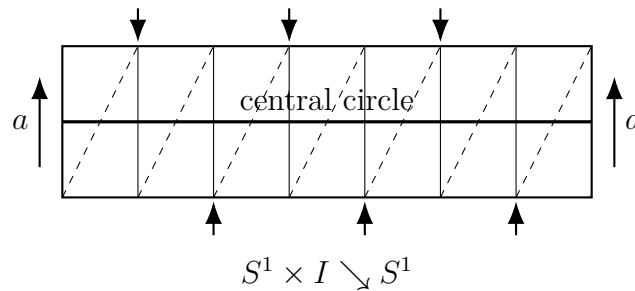
Each vertex belongs to two edges, so no vertex is free.

Therefore the circle cannot collapse to a point. Its homology is

$$H_0(S^1) \cong \mathbb{Z}, \quad H_1(S^1) \cong \mathbb{Z}.$$

6.7 Diagram 7: Annulus collapses to its central circle

An annulus can be triangulated as a strip whose left and right sides are identified.

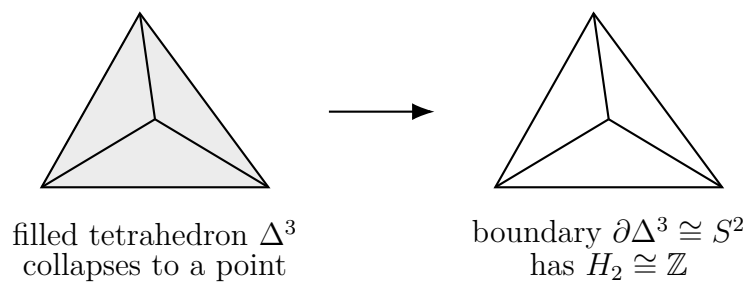


The annulus does not collapse to a point, because the central circle represents a genuine H_1 -class:

$$H_1(S^1 \times I) \cong \mathbb{Z}.$$

6.8 Diagram 8: Filled tetrahedron versus boundary sphere

A filled tetrahedron Δ^3 is collapsible. Its boundary $\partial\Delta^3 \cong S^2$ is not collapsible to a point.



The filled tetrahedron has

$$H_0 \cong \mathbb{Z}, \quad H_j = 0 \quad j \geq 1.$$

The boundary sphere has

$$H_0(S^2) \cong \mathbb{Z}, \quad H_2(S^2) \cong \mathbb{Z}.$$

6.9 Diagram 9: Chain cancellation picture

An elementary collapse removes an acyclic two-term chain piece:

$$\mathbb{Z}[\sigma] \xrightarrow{\cong} \mathbb{Z}[\tau].$$

$$0 \longrightarrow \mathbb{Z}[\sigma] \xrightarrow{\partial} \mathbb{Z}[\tau] \longrightarrow 0$$

$$\sigma \longmapsto \tau$$

Since this relative complex has zero homology, removing σ and τ preserves the homology of the whole complex.

6.10 Chart 1: What collapses and what does not collapse

Complex	Free face exists?	Collapse result	Homology
interval	yes	point	$H_0 \cong \mathbb{Z}$
filled triangle	yes	point	$H_0 \cong \mathbb{Z}$
tree	yes, leaf vertices	point	$H_0 \cong \mathbb{Z}$
circle	no	does not collapse to point	$H_0 \cong \mathbb{Z}, H_1 \cong \mathbb{Z}$
annulus	yes on boundary	S^1	$H_0 \cong \mathbb{Z}, H_1 \cong \mathbb{Z}$
filled tetrahedron	yes	point	$H_0 \cong \mathbb{Z}$
$\partial\Delta^3 \cong S^2$	no 2-dimensional free face	does not collapse to point	$H_0 \cong \mathbb{Z}, H_2 \cong \mathbb{Z}$
T^2	no collapse to point	obstructed by homology	$H_0 \cong \mathbb{Z}, H_1 \cong \mathbb{Z}^2, H_2 \cong \mathbb{Z}$

6.11 Chart 2: Collapse versus homology

Situation	Geometric meaning	Homological meaning
free face exists	a simplex is attached in a removable way	acyclic cancellation pair
$K \searrow L$	remove (σ, τ)	$H_j(K) \cong H_j(L)$
$K \searrow \text{point}$	complex is collapsible	$H_0 \cong \mathbb{Z}, H_j = 0 \text{ for } j \geq 1$
no free face	nothing removable by collapse	may contain genuine cycles
$H_1 \neq 0$	loop exists	cannot collapse to a point
$H_2 \neq 0$	closed surface exists	cannot collapse to a point

6.12 Final visual summary

$$\begin{array}{ccc}
 K & \xrightarrow{\text{remove free pair } (\sigma, \tau)} & L = K \setminus \{\sigma, \tau\} \\
 & \searrow H_j(K) & \downarrow H_j(L) \\
 & & H_j(K) \cong H_j(L)
 \end{array}$$

The essential idea is:

Elementary collapse removes an acyclic piece, so homology is unchanged.

7 Deep theory for Problems 4–5: connected sums, intersection forms, and parity phenomena

7.1 Problem 4: same homology, not homotopy equivalent

We consider the closed oriented 4-manifolds

$$X_1 = \mathbb{CP}^2 \# \mathbb{CP}^2, \quad X_2 = \mathbb{CP}^2 \# \overline{\mathbb{CP}^2}, \quad X_3 = S^2 \times S^2,$$

and we want to understand why

$$H_*(X_1) \cong H_*(X_2) \cong H_*(X_3)$$

as graded abelian groups, while no two of the X_i are homotopy equivalent.

A. Homology of a connected sum in dimension $n \geq 3$

Let M and N be closed connected n -manifolds with $n \geq 3$. Remove the interior of an embedded n -ball from each and glue along the boundary sphere:

$$M \# N := (M \setminus \text{int } D^n) \cup_{S^{n-1}} (N \setminus \text{int } D^n).$$

Write $M^\circ := M \setminus \text{int } D^n$ and $N^\circ := N \setminus \text{int } D^n$, so that

$$M \# N = M^\circ \cup N^\circ, \quad M^\circ \cap N^\circ \cong S^{n-1}.$$

Applying Mayer–Vietoris to (M°, N°) and using $H_i(S^{n-1}) = 0$ for $0 < i < n - 1$ yields the standard conclusion:

$$H_i(M \# N) \cong H_i(M) \oplus H_i(N) \quad \text{for } 1 \leq i \leq n - 2.$$

In the top degrees one still has

$$H_0(M \# N) \cong \mathbb{Z}, \quad H_n(M \# N) \cong \mathbb{Z},$$

and the behavior in degree $n - 1$ can be analyzed similarly (in our applications below, $n = 4$ so $n - 1 = 3$ and the groups are typically 0 anyway).

B. Apply to X_1, X_2, X_3

Recall:

$$H_*(\mathbb{CP}^2) : \quad H_0 \cong \mathbb{Z}, \quad H_2 \cong \mathbb{Z}, \quad H_4 \cong \mathbb{Z}, \quad \text{others } 0,$$

and orientation reversal does not change homology groups:

$$H_*(\overline{\mathbb{CP}^2}) \cong H_*(\mathbb{CP}^2).$$

Also

$$H_*(S^2 \times S^2) : \quad H_0 \cong \mathbb{Z}, \quad H_2 \cong \mathbb{Z}^2, \quad H_4 \cong \mathbb{Z}, \quad \text{others } 0.$$

Using the connected-sum homology rule for $n = 4$ (so for $i = 1, 2$ we get direct sums, and in these examples $H_1 = H_3 = 0$), we obtain:

$$H_2(\mathbb{CP}^2 \# \mathbb{CP}^2) \cong \mathbb{Z} \oplus \mathbb{Z} \cong \mathbb{Z}^2,$$

$$H_2(\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}) \cong \mathbb{Z} \oplus \mathbb{Z} \cong \mathbb{Z}^2,$$

and of course $H_0 \cong \mathbb{Z}$, $H_4 \cong \mathbb{Z}$, and $H_1 = H_3 = 0$ for all three. Hence

$$H_*(X_1) \cong H_*(X_2) \cong H_*(X_3)$$

as graded abelian groups.

C. Why homology does *not* determine homotopy type in dimension 4

For a closed oriented 4-manifold X , the key additional invariant is the *intersection form* on middle cohomology:

$$Q_X : H^2(X; \mathbb{Z})/\text{tors} \times H^2(X; \mathbb{Z})/\text{tors} \longrightarrow \mathbb{Z}, \quad Q_X(\alpha, \beta) = \langle \alpha \smile \beta, [X] \rangle.$$

This is a unimodular symmetric bilinear form (unimodular meaning the induced map to the dual lattice is an isomorphism).

Remark 7.1 (Homotopy invariance). A homotopy equivalence $f : X \rightarrow Y$ induces a ring isomorphism

$$f^* : H^*(Y; \mathbb{Z}) \xrightarrow{\cong} H^*(X; \mathbb{Z}),$$

and it sends $[Y]$ to $\pm[X]$ in top degree. Therefore f^* identifies the cup product pairings, and in particular identifies the intersection forms up to change of $\mathbb{Z}$ -basis (integral congruence). Thus:

$$X \simeq Y \implies Q_X \cong Q_Y \text{ (as integral forms, up to sign depending on orientation).}$$

D. Behavior under connected sum and orientation reversal

Two fundamental structural facts:

- **(Connected sum)** For oriented 4-manifolds,

$$Q_{X \# Y} \cong Q_X \oplus Q_Y.$$

- **(Orientation reversal)** If $\overline{X}$ is X with reversed orientation, then

$$Q_{\overline{X}} \cong -Q_X.$$

E. Compute Q for the three examples

1. The form of $\mathbb{CP}^2$. Let $a \in H^2(\mathbb{CP}^2; \mathbb{Z})$ be the standard generator (hyperplane class). Then

$$\langle a \smile a, [\mathbb{CP}^2] \rangle = 1,$$

so

$$Q_{\mathbb{CP}^2} = [1].$$

Hence

$$Q_{\overline{\mathbb{CP}^2}} = [-1].$$

2. The form of $X_1 = \mathbb{CP}^2 \# \mathbb{CP}^2$. By direct sum:

$$Q_{X_1} \cong [1] \oplus [1].$$

This form is *positive definite* and has signature $+2$.

3. The form of $X_2 = \mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$.

$$Q_{X_2} \cong [1] \oplus [-1].$$

This form is *indefinite* with signature 0, and it is *odd* (since it has a vector of square 1).

4. The form of $X_3 = S^2 \times S^2$. Let $u = [S^2 \times \{*\}]^\vee$ and $v = [\{*\} \times S^2]^\vee$ be the standard basis of $H^2(S^2 \times S^2; \mathbb{Z}) \cong \mathbb{Z}^2$. Then

$$u \smile u = 0, \quad v \smile v = 0, \quad u \smile v \text{ evaluates to 1 on } [S^2 \times S^2],$$

so in the basis (u, v) we have

$$Q_{S^2 \times S^2} \cong \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This form is *indefinite* with signature 0, and it is *even* (every vector has even self-intersection).

F. Separate the homotopy types

We now compare invariants of the forms:

- Q_{X_1} is *definite* (signature $+2$), while Q_{X_2} and Q_{X_3} are *indefinite* (signature 0). Therefore X_1 is not homotopy equivalent to X_2 or X_3 .
- Q_{X_2} is *odd*, while Q_{X_3} is *even*. Therefore X_2 is not homotopy equivalent to X_3 .

Hence no two of X_1, X_2, X_3 are homotopy equivalent, despite having the same homology groups.

Remark 7.2 (Takeaway). Homology groups alone do not determine the homotopy type of a closed oriented 4-manifold. The *cup product structure* in degree 2 (equivalently, the intersection form) is a strictly finer invariant and is preserved by homotopy equivalence.

7.2 Problem 5: parity of the middle Betti number in dimension $4n + 2$

Let M be a closed $\mathbb{Z}$ -orientable manifold of dimension $4n + 2$. We show that

$$\dim_{\mathbb{Q}} H_{2n+1}(M; \mathbb{Q}) \text{ is even.}$$

A. Poincaré duality gives a perfect middle pairing

Since M is $\mathbb{Z}$ -orientable, it is also $\mathbb{Q}$ -orientable, and Poincaré duality gives a perfect pairing

$$H^k(M; \mathbb{Q}) \times H^{(4n+2)-k}(M; \mathbb{Q}) \longrightarrow \mathbb{Q}, \quad (\alpha, \beta) \mapsto \langle \alpha \smile \beta, [M] \rangle.$$

In particular, when $k = 2n + 1$ we obtain a bilinear pairing on the *same* vector space:

$$\langle \cdot, \cdot \rangle : H^{2n+1}(M; \mathbb{Q}) \times H^{2n+1}(M; \mathbb{Q}) \rightarrow \mathbb{Q}, \quad \langle \alpha, \beta \rangle = \langle \alpha \smile \beta, [M] \rangle,$$

and Poincaré duality implies this pairing is nondegenerate.

B. Graded commutativity forces skew-symmetry

The cup product is graded-commutative:

$$\alpha \smile \beta = (-1)^{|\alpha||\beta|} \beta \smile \alpha.$$

Here $|\alpha| = |\beta| = 2n + 1$ is odd, hence $|\alpha||\beta|$ is odd and

$$\alpha \smile \beta = -\beta \smile \alpha.$$

Therefore the pairing satisfies

$$\langle \alpha, \beta \rangle = -\langle \beta, \alpha \rangle,$$

so it is *skew-symmetric*. Over $\mathbb{Q}$ (characteristic $\neq 2$), skew-symmetry implies

$$\langle \alpha, \alpha \rangle = 0 \quad \text{for all } \alpha.$$

Thus the pairing is an *alternating* nondegenerate bilinear form.

C. Linear algebra lemma: nondegenerate alternating forms have even dimension

Lemma 7.1 (Symplectic dimension parity). Let V be a finite-dimensional vector space over a field of characteristic $\neq 2$, and let $\omega : V \times V \rightarrow \mathbb{k}$ be a nondegenerate alternating bilinear form. Then $\dim V$ is even.

Proof. Over such a field, one can build a symplectic basis by induction: choose $v \neq 0$, then nondegeneracy gives w with $\omega(v, w) \neq 0$, and $\langle v, w \rangle$ is a 2-dimensional nondegenerate subspace; take its orthogonal complement and continue. Hence V decomposes into a direct sum of 2-dimensional symplectic blocks, so $\dim V$ is even. $\square$

Applying the lemma to

$$V = H^{2n+1}(M; \mathbb{Q})$$

with $\omega(\alpha, \beta) = \langle \alpha \smile \beta, [M] \rangle$ yields:

$$\dim_{\mathbb{Q}} H^{2n+1}(M; \mathbb{Q}) \text{ is even.}$$

D. Convert to homology

Over a field, universal coefficients gives

$$H^{2n+1}(M; \mathbb{Q}) \cong \text{Hom}(H_{2n+1}(M; \mathbb{Q}), \mathbb{Q}),$$

so

$$\dim_{\mathbb{Q}} H^{2n+1}(M; \mathbb{Q}) = \dim_{\mathbb{Q}} H_{2n+1}(M; \mathbb{Q}).$$

Therefore $\dim_{\mathbb{Q}} H_{2n+1}(M; \mathbb{Q})$ is even, as desired.

Remark 7.3 (Interpretation). In dimension $4n + 2$, the middle cup-product pairing is symplectic (alternating) rather than symmetric, forcing the middle Betti number to come in pairs.

7.3 Examples for Problem 5: evenness of the middle Betti number in dimension $4n + 2$

Recall the statement: if M is a closed $\mathbb{Z}$ -orientable manifold of dimension $4n + 2$, then

$$\dim_{\mathbb{Q}} H_{2n+1}(M; \mathbb{Q}) \text{ is even.}$$

Equivalently, the middle Betti number $b_{2n+1}(M)$ is even (over $\mathbb{Q}$).

Example 1: Closed oriented surfaces Σ_g (the case $n = 0$)

Here $\dim \Sigma_g = 2 = 4 \cdot 0 + 2$, so the relevant group is $H_1(\Sigma_g; \mathbb{Q})$. We have the classical computation:

$$H_0(\Sigma_g; \mathbb{Q}) \cong \mathbb{Q}, \quad H_1(\Sigma_g; \mathbb{Q}) \cong \mathbb{Q}^{2g}, \quad H_2(\Sigma_g; \mathbb{Q}) \cong \mathbb{Q}.$$

Hence

$$\dim_{\mathbb{Q}} H_1(\Sigma_g; \mathbb{Q}) = 2g,$$

which is even, as predicted. The cup product pairing on $H^1(\Sigma_g; \mathbb{Q})$ is the standard nondegenerate skew-symmetric (symplectic) intersection pairing.

Example 2: $\Sigma_g \times S^4$ (dimension 6, i.e. $n = 1$)

Let

$$M = \Sigma_g \times S^4.$$

Then $\dim M = 2 + 4 = 6 = 4 \cdot 1 + 2$, so the middle degree is $2n + 1 = 3$. Using Künneth over $\mathbb{Q}$,

$$H_3(\Sigma_g \times S^4; \mathbb{Q}) \cong H_3(\Sigma_g; \mathbb{Q}) \otimes H_0(S^4; \mathbb{Q}) \oplus H_1(\Sigma_g; \mathbb{Q}) \otimes H_2(S^4; \mathbb{Q}) \oplus H_0(\Sigma_g; \mathbb{Q}) \otimes H_3(S^4; \mathbb{Q}).$$

But

$$H_3(\Sigma_g; \mathbb{Q}) = 0, \quad H_2(S^4; \mathbb{Q}) = 0, \quad H_3(S^4; \mathbb{Q}) = 0,$$

so

$$H_3(\Sigma_g \times S^4; \mathbb{Q}) = 0,$$

and therefore $\dim H_3 = 0$, which is even.

Example 3: $S^3 \times S^3$ (dimension 6, i.e. $n = 1$)

Let

$$M = S^3 \times S^3.$$

Then $\dim M = 6$ and the middle degree is 3. By Künneth over $\mathbb{Q}$,

$$H_3(S^3 \times S^3; \mathbb{Q}) \cong H_3(S^3; \mathbb{Q}) \otimes H_0(S^3; \mathbb{Q}) \oplus H_0(S^3; \mathbb{Q}) \otimes H_3(S^3; \mathbb{Q}) \cong \mathbb{Q} \oplus \mathbb{Q}.$$

Hence

$$\dim_{\mathbb{Q}} H_3(S^3 \times S^3; \mathbb{Q}) = 2,$$

which is even. In this example the middle cohomology pairing is a 2×2 skew form.

Example 4: Connected sums $\#^k(S^3 \times S^3)$ (dimension 6)

Let

$$M = \#^k(S^3 \times S^3).$$

Then M is closed oriented of dimension 6, and the middle degree is 3. For connected sums in dimension ≥ 3 ,

$$H_3(\#^k(S^3 \times S^3); \mathbb{Q}) \cong \bigoplus_{j=1}^k H_3(S^3 \times S^3; \mathbb{Q}) \cong (\mathbb{Q}^2)^{\oplus k} \cong \mathbb{Q}^{2k}.$$

Thus

$$\dim_{\mathbb{Q}} H_3(M; \mathbb{Q}) = 2k,$$

again even. This gives a large family where the middle Betti number can be any even integer.

Example 5: Products $\Sigma_g \times S^{4n}$ (general $n \geq 1$)

Let $n \geq 1$ and set

$$M = \Sigma_g \times S^{4n}.$$

Then $\dim M = 2 + 4n = 4n + 2$, and the middle degree is $2n + 1$. By Künneth over $\mathbb{Q}$,

$$H_{2n+1}(M; \mathbb{Q}) \cong \bigoplus_{i+j=2n+1} H_i(\Sigma_g; \mathbb{Q}) \otimes H_j(S^{4n}; \mathbb{Q}).$$

But the only nonzero rational homology of S^{4n} is in degrees 0 and $4n$, so the only possible summands would require $j = 0$ or $j = 4n$, i.e.

$$i = 2n + 1 \quad \text{or} \quad i = 2n + 1 - 4n = 1 - 2n.$$

For $n \geq 1$, both indices are impossible for Σ_g (the first is > 2 , the second is negative), so

$$H_{2n+1}(\Sigma_g \times S^{4n}; \mathbb{Q}) = 0.$$

Hence the middle Betti number is 0, which is even.

Remark: why orientability matters and why $\mathbb{F}_2$ is different

The evenness argument uses a nondegenerate *alternating* pairing on $H^{2n+1}(M; \mathbb{Q})$. Over a field of characteristic 2, “skew-symmetric” and “symmetric” coincide, and one can no longer conclude that the dimension must be even. This is why the clean parity statement is formulated over $\mathbb{Q}$ (or any field of characteristic $\neq 2$).

7.4 Closed and orientable vs. $\mathbb{Z}$ -orientable and $\mathbb{Q}$ -orientable**Closed manifolds**

An n -manifold M is **closed** if it is *compact* and has *empty boundary*:

$$M \text{ closed} \iff M \text{ compact and } \partial M = \emptyset.$$

Examples: S^n , T^n , $\mathbb{CP}^m$. Non-examples: D^n (has boundary), $\mathbb{R}^n$ (not compact).

Orientable manifolds

A manifold M is **orientable** if one can choose a consistent orientation on all coordinate charts (equivalently, all transition maps between oriented charts have positive Jacobian determinant in the smooth case). There are several equivalent algebraic-topological characterizations.

Remark 7.4 (Homology characterization for closed connected manifolds). If M is a connected closed n -manifold, then

$$M \text{ orientable} \iff H_n(M; \mathbb{Z}) \cong \mathbb{Z},$$

and

$$M \text{ nonorientable} \iff H_n(M; \mathbb{Z}) = 0.$$

 $\mathbb{Z}$ -orientable

Saying that M is **$\mathbb{Z}$ -orientable** means that M admits a fundamental class with integer coefficients. For a connected closed n -manifold this is exactly the usual notion of orientability:

$$\mathbb{Z}\text{-orientable} \iff \text{orientable}.$$

 F -orientable and $\mathbb{Q}$ -orientable

Let F be a field. A connected closed n -manifold M is called **F -orientable** if it admits a fundamental class $[M]_F \in H_n(M; F)$, equivalently

$$H_n(M; F) \cong F.$$

In particular, M is **$\mathbb{Q}$ -orientable** if

$$H_n(M; \mathbb{Q}) \cong \mathbb{Q}.$$

Remark 7.5 (Relations between $\mathbb{Z}$ - and $\mathbb{Q}$ -orientability). For closed connected manifolds one has

$$\mathbb{Z}\text{-orientable} \implies \mathbb{Q}\text{-orientable}.$$

Moreover, since $\mathbb{Q}$ has characteristic $\neq 2$, for manifolds one actually has the equivalence

$$\mathbb{Q}\text{-orientable} \iff \mathbb{Z}\text{-orientable} \iff \text{orientable}.$$

The special role of $\mathbb{F}_2$

Over the field $\mathbb{F}_2$, signs disappear ($-1 = +1$), and every manifold becomes orientable.

Remark 7.6 ($\mathbb{F}_2$ -orientability). Every connected closed manifold M is $\mathbb{F}_2$ -orientable:

$$H_n(M; \mathbb{F}_2) \cong \mathbb{F}_2.$$

For example, $\mathbb{RP}^2$ is not $\mathbb{Z}$ -orientable (so $H_2(\mathbb{RP}^2; \mathbb{Z}) = 0$), but it is $\mathbb{F}_2$ -orientable (so $H_2(\mathbb{RP}^2; \mathbb{F}_2) \cong \mathbb{F}_2$).

Why this matters for Poincaré duality

Let M be a connected closed n -manifold and let F be a field. If M is F -orientable, then Poincaré duality holds with coefficients in F :

$$H^i(M; F) \cong H_{n-i}(M; F),$$

via cap product with the fundamental class $[M]_F$. In particular:

- If M is orientable, then duality holds over *any* field F (including $\mathbb{Q}$).
- For an arbitrary (possibly nonorientable) manifold, duality always holds over $\mathbb{F}_2$.

Summary

For connected closed manifolds:

$$\text{orientable} \iff \mathbb{Z}\text{-orientable} \iff \mathbb{Q}\text{-orientable},$$

while

$$\text{every manifold is } \mathbb{F}_2\text{-orientable}.$$